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Quantum Computing for Ubiquitous and Pervasive Computing Systems

Moustafa Youssef, Wei Li, Rahul Yadav, Muhammad Aleem Raza

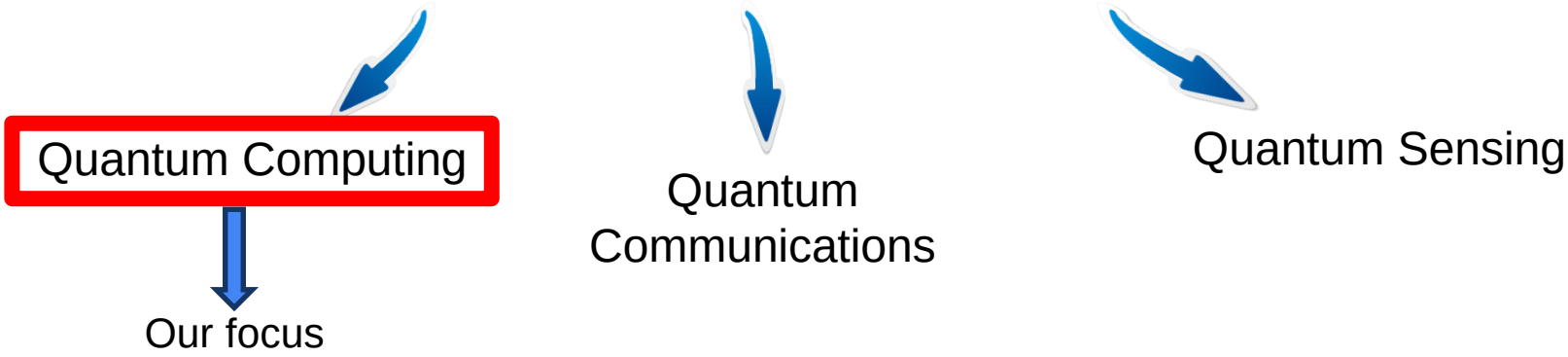
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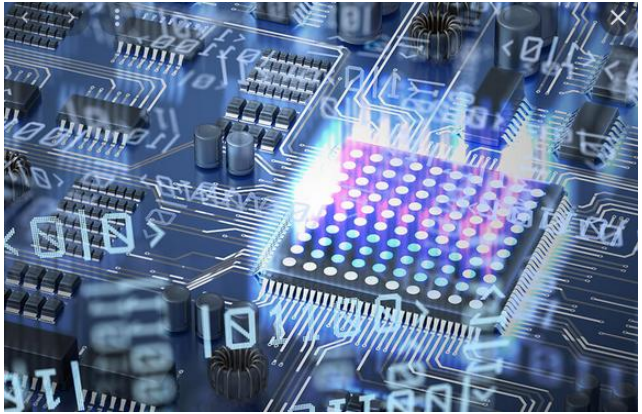
Major Digital Quantum Technologies



Quantum Computing

- A new field at the intersection of physics, mathematics, and computer science

- Leverages the phenomena of quantum mechanics to improve the efficiency of computation
  - Superposition, Interference, Entanglement, etc



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Quantum Computing



Quantum Circuit Model
or Quantum Gate Model



Quantum Annealing

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Quantum Computing in the News

- Shor’s algorithm has put our current encryption methods at risk
- Quantum key distribution will make encryption safe again
- Grover’s algorithm will speed up data searches
- China teleported a qubit from earth to a satellite
- Google claims quantum supremacy
  - IBM argues that this is not true
- [Oct 2025]Google has a useful quantum algorithm that outperforms a supercomputer
  - An approach it calls “quantum echoes” takes 13,000 times longer on a supercomputer.

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Quantum Computing in the News

- “We don't need any more fundamental breakthroughs. We need little improvements here and there...We have all the pieces together. We just need to integrate them well to build larger and larger systems."
- "[Google’s Hartmut] Neven believes this can be accomplished by the end of the decade. IBM's Gil also predicts around the same timeline."

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Quantum Computing Echo System 2024



Src: McKinsey & Company

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Quantum Computing Echo System 2024

- McKinsey research has found
  - There is only **one qualified quantum candidate for every three quantum job** openings
  - By 2025, predicts that **< 50 percent of quantum jobs will be filled**

- Time is **NOW**
  - To invest in QC
    - Personally and as instutuions

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What's Quantum Computing

- Fusion of Quantum Physics and Computer Science
  - Entirely new way of thinking about computation
  - Efficiently solves (and breaks) hard problems for classical computers
    - Encryption, search, optimization, others

- Basic unit of computation: **QuBit**

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QuBit

- Similarities to the classical bit:
  - Basic states: takes either 0 or 1
- Differences from the classical bit:
  - Superposition
  - Measurement (reading the bit)
  - Entanglement

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Qubit - Superposition



- Quantum Bit
- Classical bits: 0 or 1
  - If you measure 0, you will get 0 and vice versa
- Qubit:
  - Can be in an **infinite** number of states!
    - Superposition of **BOTH** 0 and 1
    - Leverage this to speed up computations
      - Quantum parallelism



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Qubit - Measurement

- Qubit:
  - Can be in an **infinite** number of states!
    - Superposition of **BOTH** 0 and 1
- When we measure it
  - We get a 0 or 1
    - Leverage this to get the final answer
  - The act of measuring affects the quantum bit
  - A SIMPLE mathematical model captures this!

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Qubit - Entanglement

- Qubits can also be entangled:
  - When we make a measurement of one of them, it **affects the state of the other**
  - Regardless of the **distance** between them!
  - Einstein: “Spooky action at distance” ☺
- Described perfectly by our simple mathematical model

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Quantum Computing

- Leverages Superposition, Entanglement, and Measurement
  - Key Quantum Mechanics concepts

+

- Human ingenuity

- To devise new algorithms that perform significantly better than classical computers

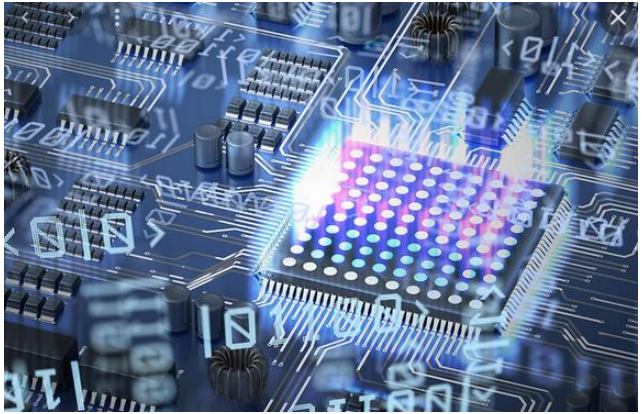
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Quantum Computing vs Classical Computing

- Two distinct disciplines, but
- Quantum computing is the more fundamental form of computing:
  - Anything that can be computed classically can be computed on a quantum computer
- The qubit is the basic unit of computation
  - Not the bit
- Computation, in its essence, really means quantum computation
- This is how nature behaves!

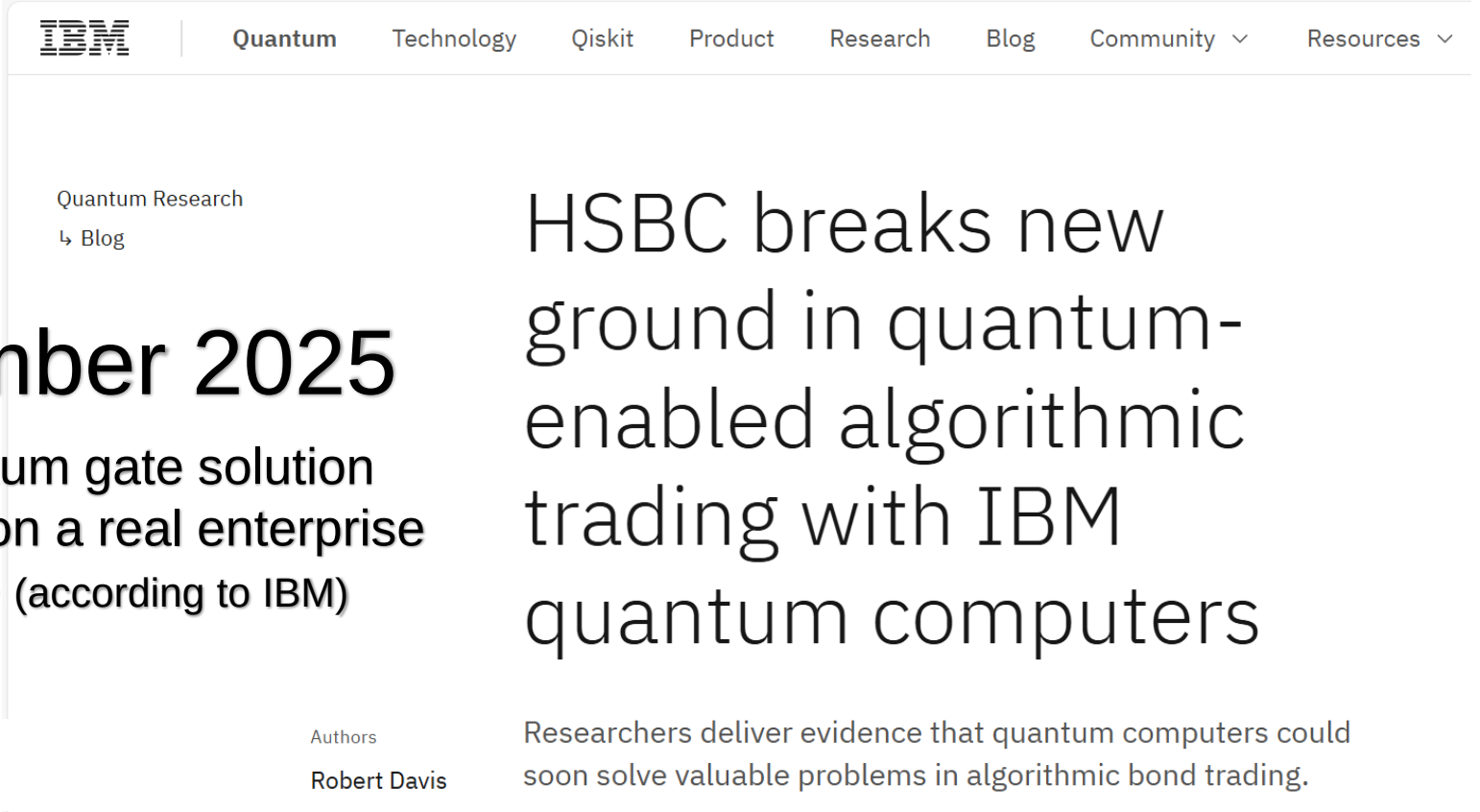
Quantum Computing – Why now?

- Growing advancements
  - in both quantum hardware and software
- More promising potential over time
- Big investments
  - Industry and private sector



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Recent Quantum News



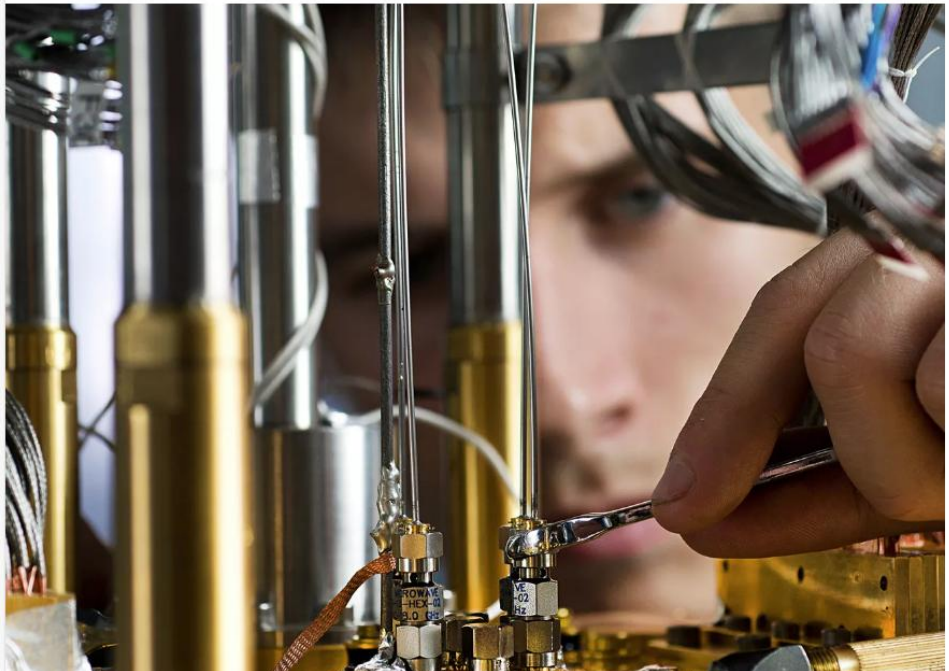
September 2025

First quantum gate solution deployment on a real enterprise use case (according to IBM)

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Quantum Annealing in Enterprise

Menu



Future of Mobility

Volkswagen takes the quantum computing revolution from the lab to the factory

July 15, 2021

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Recent Quantum Investments

State of Maryland, University of Maryland Announce $1 Billion "Capital of Quantum" Initiative

January 14, 2025

UK And US Seal Tech Pact With £31 Billion AI And Quantum Push

National, Quantum Computing Business, Ukquantum Matt Swayne • September 17, 2025

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Recent Quantum Investments

- China's 15th Five-Year Plan (2026-2030), adopted in March, names quantum technology **as the first of seven "future industries"** earmarked as new engines of economic growth.
- China's government investment in quantum technology is widely cited at approximately $15 billion.
- China Telecom Latest Report Flags Quantum as a Key Growth Driver (65.4% revenue growth in the Quantum Sector)

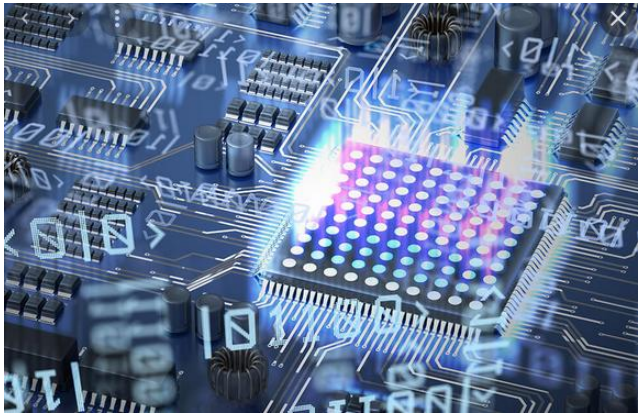
  - Src: https://thequantuminsider.com/2026/04/06/chinas-quantum-sector-sees-investment-surge-as-larger-funding-rounds-return/
  - https://thequantuminsider.com/2026/03/27/china-telecom-latest-report-flags-quantum-as-a-key-growth-driver-but-scale-remains-uncertain/

Quantum Computing – Why now?

- Practical algorithms
  - Shor’s factoring algorithm
  - Grover’s algorithm

- Several institutions claimed quantum supremacy

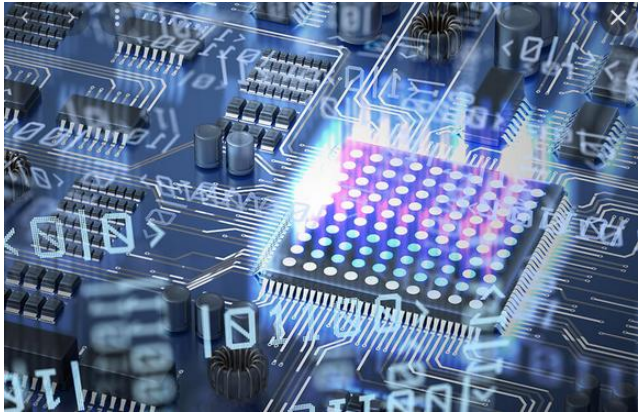
- One might expect that **quantum computing** would be a **mainstream** technology within years



Quantum Computing

- A new field at the intersection of physics, mathematics, and computer science

- Leverages the phenomena of quantum mechanics to improve the efficiency of computation
  - Superposition, Interference, Entanglement, etc

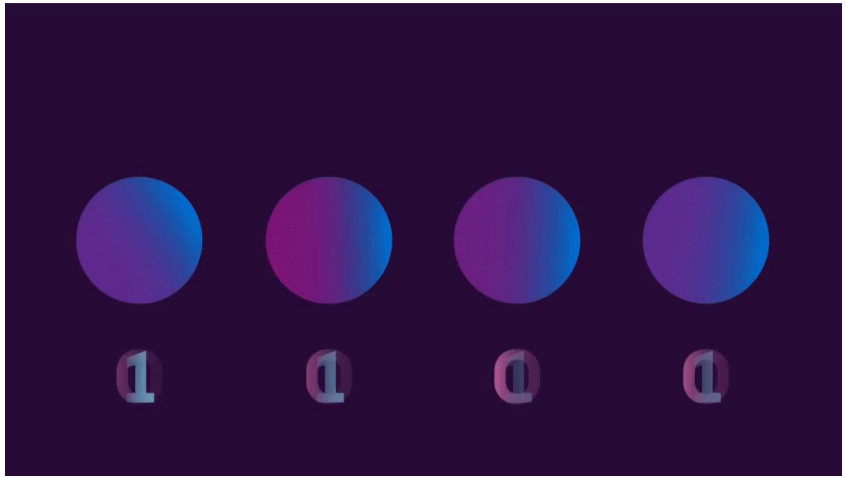


Agenda

- Introduction
- Background
- QC Algorithms Examples
- QC for Positioning and Navigation
- Conclusions and what’s next

Qubit

- Qubit is the basic unit of information.
- Qubits can have various physical **implementations**, e.g. the polarization of photons or spin of electrons
- Qubit can exist in a **superposition** of the zero and one states
- Measurement is **destructive**
  - Qubit state jumps to either |0> or |1> w.p $\boldsymbol{\alpha}^2$ and $\boldsymbol{\beta}^2$
  - Losing $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$.

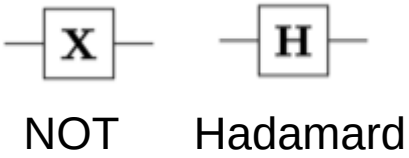


$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

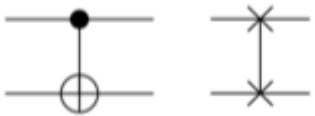
$$\alpha^2 + \beta^2 = 1$$

Quantum Gates

- Operations on qubits are usually represented by gates
  - Similar to a classical gates
- Gates can be applied to
  A single qubit
  A qubit register (group of qubits)
- Leads to changing the probability amplitudes
  - **Interference**



NOT Hadamard



CNOT SWAP

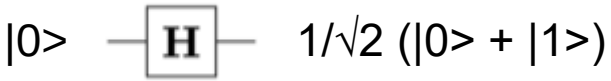
Quantum Gates – Not Gate

- Converts
  - |0> to |1>
  - |1> to |0>



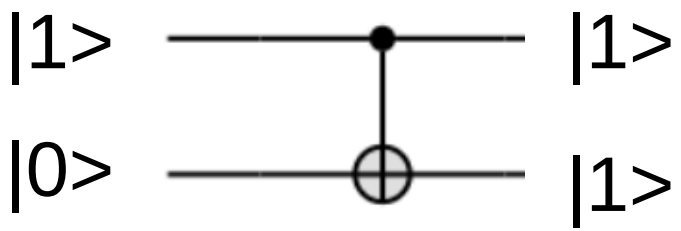
Quantum Gates - Hadamard Gate

- Converts
  - |0> to 1/√2 (|0> + |1>)
  - |1> to 1/√2 (|0> - |1>)
- Equal superposition of |0> and |1>



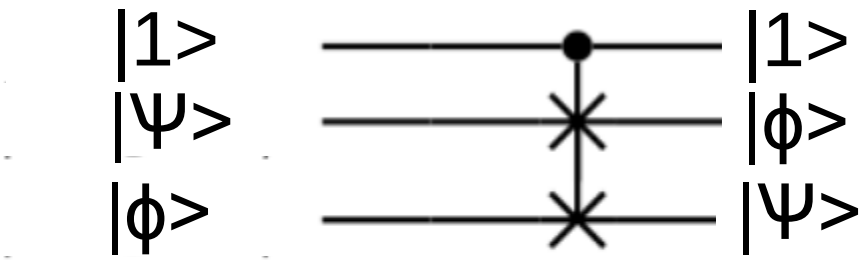
Quantum Gates – CNOT Gate

- The NOT gate is applied to the target qubit
  - If and only if, the source qubit is |1>



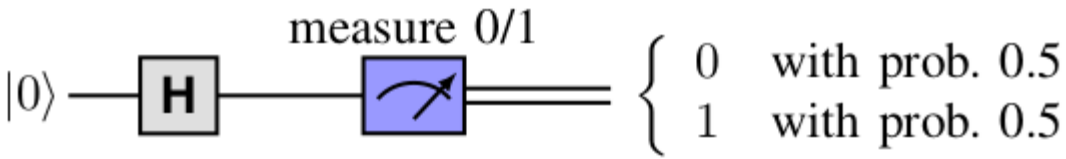
Quantum Gates – CSWAP Gate

- The two target qubits are swapped
  - If and only if, the source qubit is |1>.



Quantum Circuit

- A combination of the quantum gates.
- Single lines represent quantum information
  - Double lines represent classical information.
- The input to the circuit is a number of qubits
- The gates act on input qubits to change the combined circuit state to reach a desired output state
- The final step is to measure
  - Reveals the required informatio



Agenda

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Example of QC Algorithms

- True random number generator
- Bell’s circuit
- Quantum teleportation

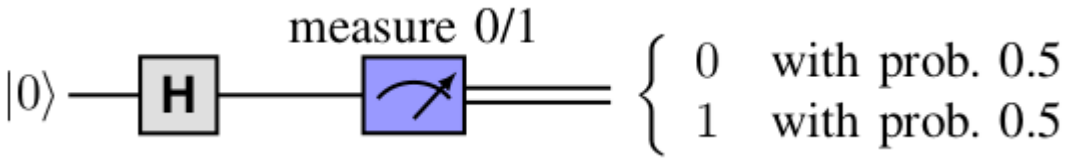
True Random Numebr Generator

Pseudo vs True Random

- rand() in C
  - Pseudo random
- QC
  - True randomness

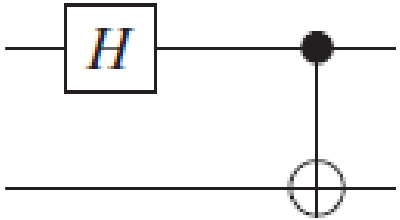
QC Random Number Generator

- Stream of 0/1’s
- Demo on Quirk



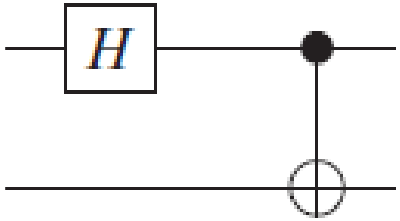
Bell’s Circuit

The Bell Circuit



- What happens when we input |00>?

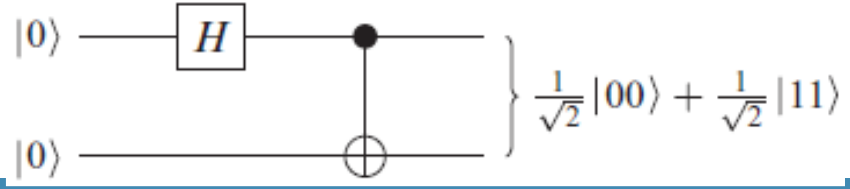
The Bell Circuit



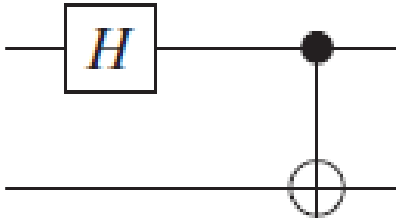
- What happens when we input |00>?
- H(|0>) = $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$
- → After H gate, the combined state is

$$\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)|0\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

- After CNOT $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$



The Bell Circuit

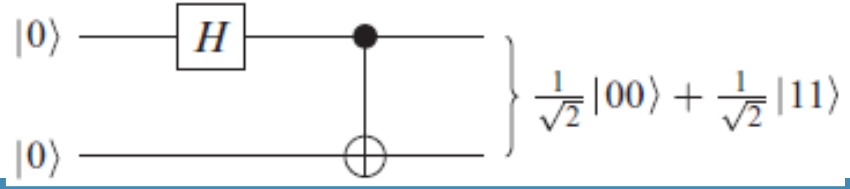


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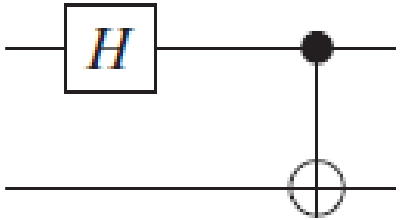
$$\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right)|0\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

- After CNOT $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$

B(|00>)= $\frac{1}{\sqrt{2}}$ |00> + $\frac{1}{\sqrt{2}}$ |11>



The Bell Circuit



- Similarly
- Each output is entangled
- What happen when we measure the first qubit?

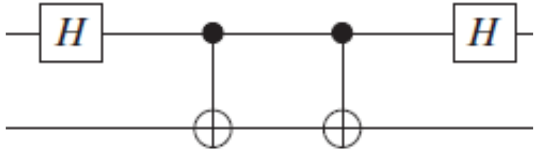
$$B(|00\rangle) = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

$$B(|01\rangle) = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

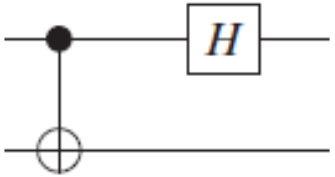
$$B(|10\rangle) = \frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle$$

$$B(|11\rangle) = \frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle$$

Example



- Output = Input
- → Reverse Bell Circuit



$$R\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle\right) = |00\rangle$$

$$R\left(\frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle\right) = |01\rangle$$

$$R\left(\frac{1}{\sqrt{2}}|00\rangle - \frac{1}{\sqrt{2}}|11\rangle\right) = |10\rangle$$

$$R\left(\frac{1}{\sqrt{2}}|01\rangle - \frac{1}{\sqrt{2}}|10\rangle\right) = |11\rangle$$

Quantum Teleportation

Note

- Sender and receiver in the same circuit

Quantum Teleportation

- Alice wants to teleport the state of her one electron to Bob
  - $a_0|0\rangle + a_1|1\rangle$
  - By sending just two classical bits
    - Can communicate infinite number of possibilities (a_0 & a_1)!
- How to do Quantum Teleportation?

Quantum Teleportation - Idea

- Use entangled electrons to control what to send
- Alice and Bob each has one electr $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$
  - In the entangled state
- Alice has another electron (to be teleported)
  - $a_0|0> + a_1|1>$
- Required:
  - Change Bob's electron (initially entangled with Alice's)
  - To be in the state $a_0|0> + a_1|1>$
- To do this, Alice will send to Bob two classical bits!

Quantum Teleportation - Idea

- Bob is going to end up with an electron in the unentangled state
  - $a_0|0\rangle + a_1|1\rangle$
- At the beginning
  - Bob's electron is entangled with Alice's
- To unentangle, you need to make a measurement
- Will Bob measure?

Quantum Teleportation - Idea

- Bob is going to end up with an electron in the unentangled state
  - $a_0|0\rangle + a_1|1\rangle$
- At the beginning
  - Bob's electron is entangled with Alice's
- To unentangle, you need to make a measurement
- Will Bob measure?
- No. o.w. he will end with a classical bit
  - → Alice will measure
  - This is how she'll get the classical bits

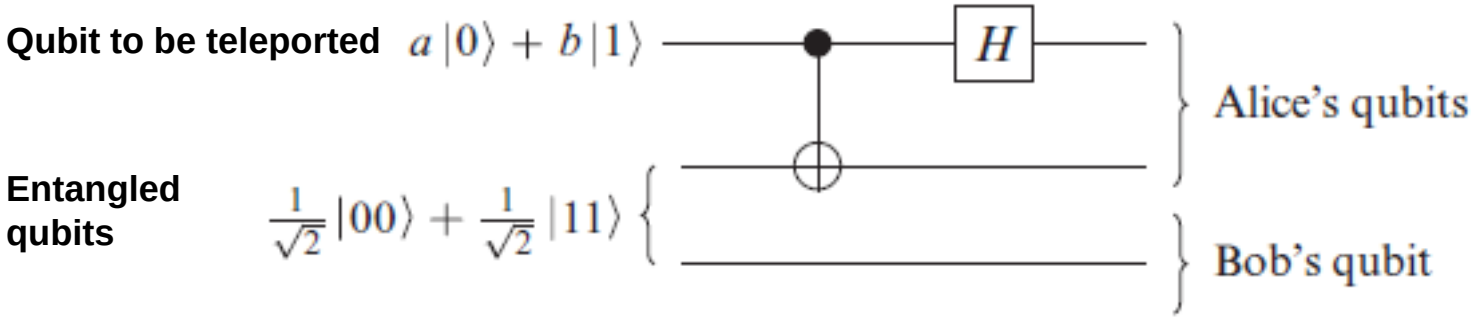
Quantum Teleportation - Idea

- How will Alice teleport the state
  - $a_0|0\rangle + a_1|1\rangle$
- To Bob?
- Need to entangle the state of this electron with the state of her other electron
  - Which is currently entangled with Bob’s
- How?

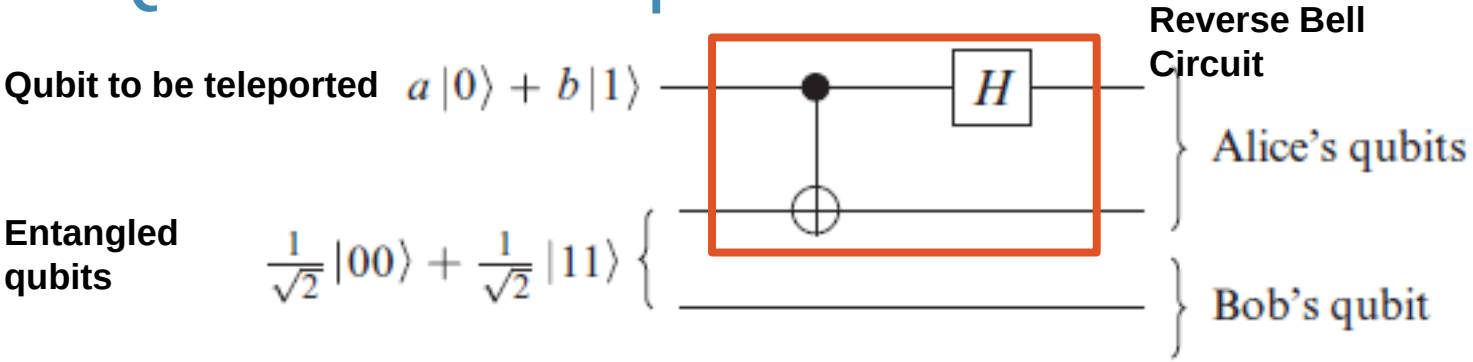
Quantum Teleportation - Idea

- How will Alice teleport the state
  - $a_0|0> + a_1|1>$
- To Bob?
- Need to entangle the state of this electron with the state of her other electron
  - Which is currently entangled with Bob's
- How?
- Obvious way:
  - Send the two qubits that she controls through a CNOT gate
- Details next

Quantum Teleportation – Full Circuit

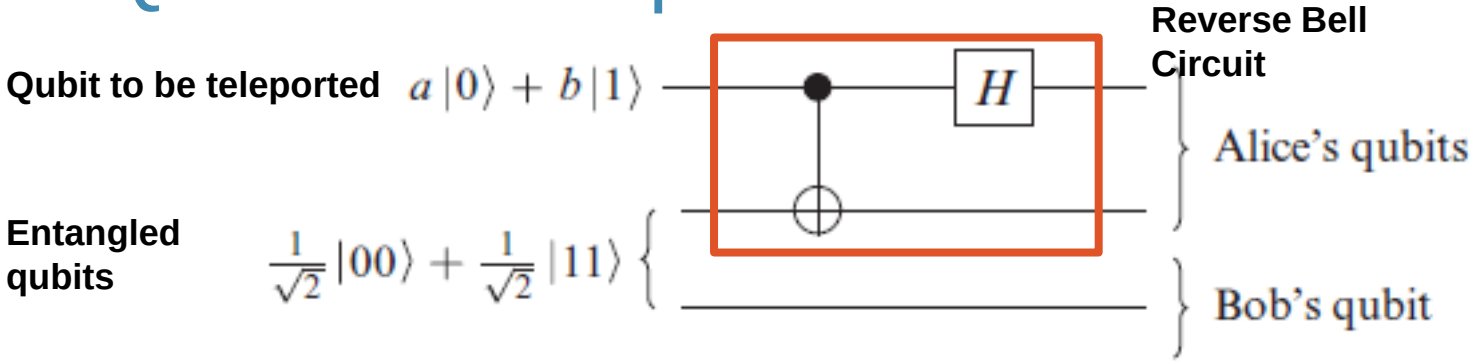


Quantum Teleportation – Full Circuit



- Initial joint state?

Quantum Teleportation – Full Circuit

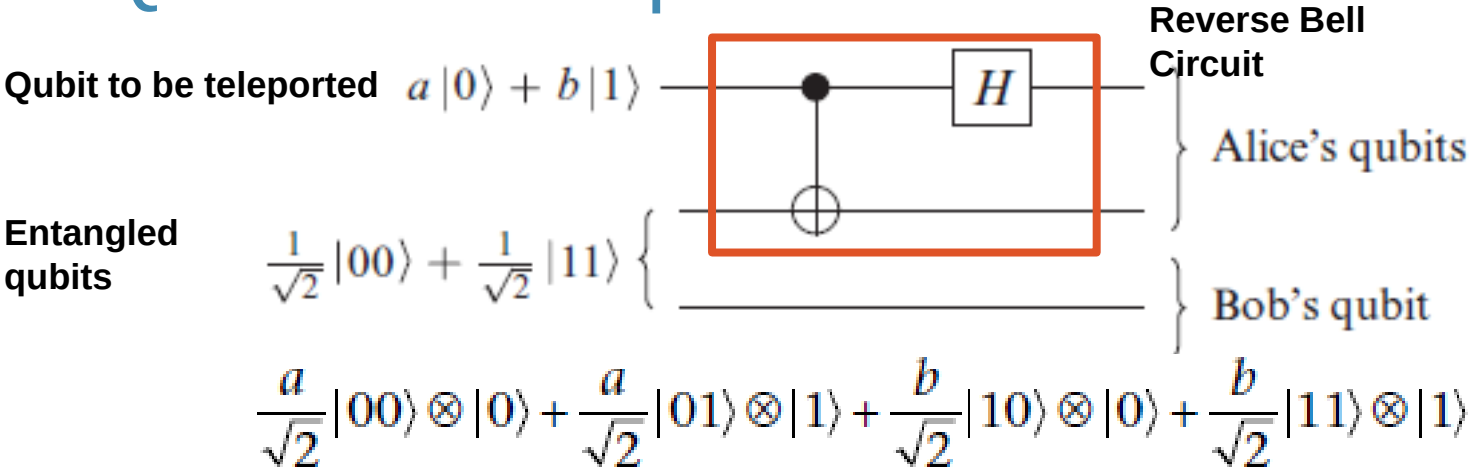


- Initial joint state?

$$(a|0\rangle + b|1\rangle) \otimes \left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle\right), \qquad \frac{a}{\sqrt{2}}|000\rangle + \frac{a}{\sqrt{2}}|011\rangle + \frac{b}{\sqrt{2}}|100\rangle + \frac{b}{\sqrt{2}}|111\rangle.$$

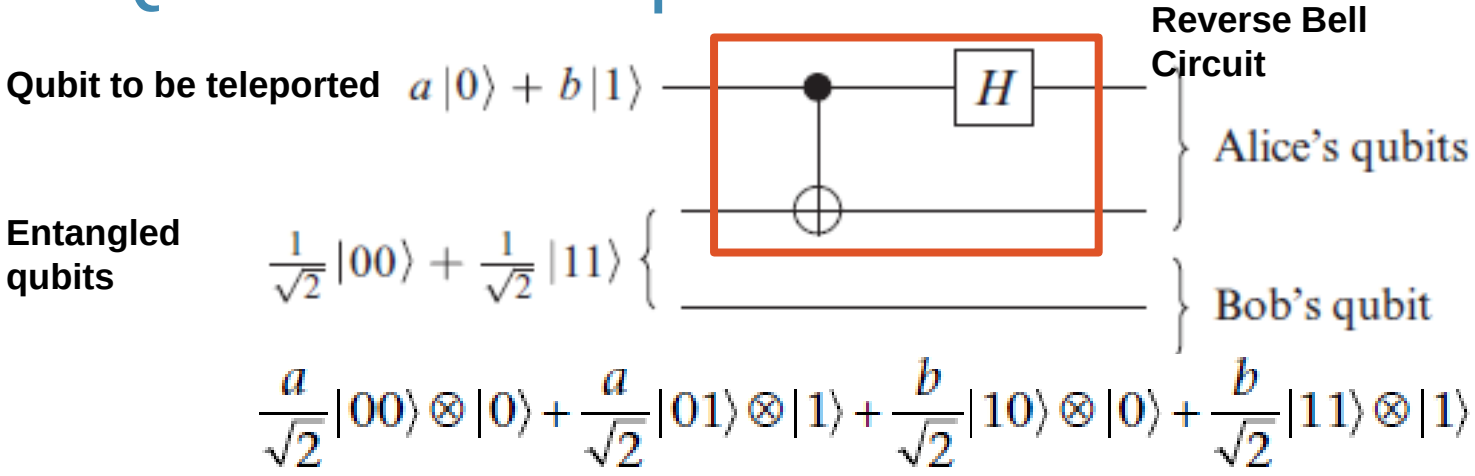
- Alice is going to work on the first 2 qubits
  - (she doesn't have the 3rd)
- Regroup $\frac{a}{\sqrt{2}}|00\rangle \otimes |0\rangle + \frac{a}{\sqrt{2}}|01\rangle \otimes |1\rangle + \frac{b}{\sqrt{2}}|10\rangle \otimes |0\rangle + \frac{b}{\sqrt{2}}|11\rangle \otimes |1\rangle$

Quantum Teleportation – Full Circuit



- **After CNOT gate?**

Quantum Teleportation – Full Circuit



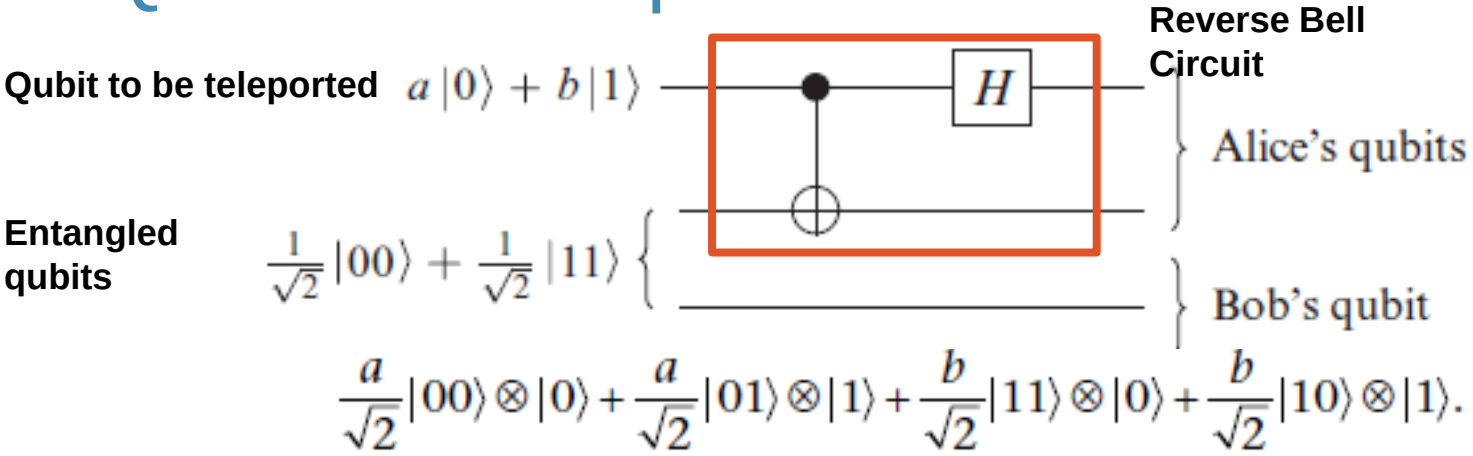
$$\frac{a}{\sqrt{2}}|00\rangle \otimes |0\rangle + \frac{a}{\sqrt{2}}|01\rangle \otimes |1\rangle + \frac{b}{\sqrt{2}}|10\rangle \otimes |0\rangle + \frac{b}{\sqrt{2}}|11\rangle \otimes |1\rangle$$

- **After CNOT gate?**

$$\frac{a}{\sqrt{2}}|00\rangle \otimes |0\rangle + \frac{a}{\sqrt{2}}|01\rangle \otimes |1\rangle + \frac{b}{\sqrt{2}}|11\rangle \otimes |0\rangle + \frac{b}{\sqrt{2}}|10\rangle \otimes |1\rangle.$$

- **After H gate?**

Quantum Teleportation – Full Circuit

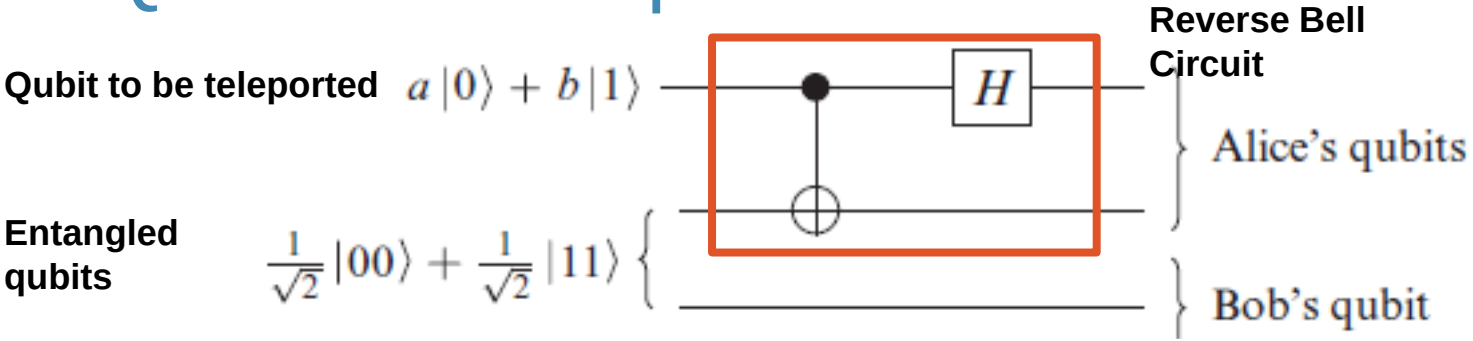


$$\frac{a}{\sqrt{2}}|00\rangle\otimes|0\rangle+\frac{a}{\sqrt{2}}|01\rangle\otimes|1\rangle+\frac{b}{\sqrt{2}}|11\rangle\otimes|0\rangle+\frac{b}{\sqrt{2}}|10\rangle\otimes|1\rangle.$$

- **After H gate?**
- **Rewrite as Alice is going to work on the 1st qubit**

$$\frac{a}{\sqrt{2}}|0\rangle\otimes|0\rangle\otimes|0\rangle+\frac{a}{\sqrt{2}}|0\rangle\otimes|1\rangle\otimes|1\rangle+\frac{b}{\sqrt{2}}|1\rangle\otimes|1\rangle\otimes|0\rangle+\frac{b}{\sqrt{2}}|1\rangle\otimes|0\rangle\otimes|1\rangle$$

Quantum Teleportation – Full Circuit



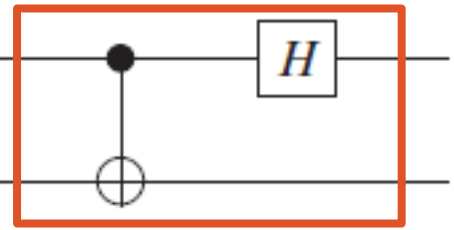
$$\frac{a}{\sqrt{2}}|0\rangle\otimes|0\rangle\otimes|0\rangle + \frac{a}{\sqrt{2}}|0\rangle\otimes|1\rangle\otimes|1\rangle + \frac{b}{\sqrt{2}}|1\rangle\otimes|1\rangle\otimes|0\rangle + \frac{b}{\sqrt{2}}|1\rangle\otimes|0\rangle\otimes|1\rangle$$

- **After H gate?**
- **H changes** $|0\rangle$ to $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ $\quad |1\rangle$ to $\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle.$

Quantum Teleportation – Full Circuit

Qubit to be teleported $a|0\rangle + b|1\rangle$

Entangled qubits $\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$



Reverse Bell Circuit

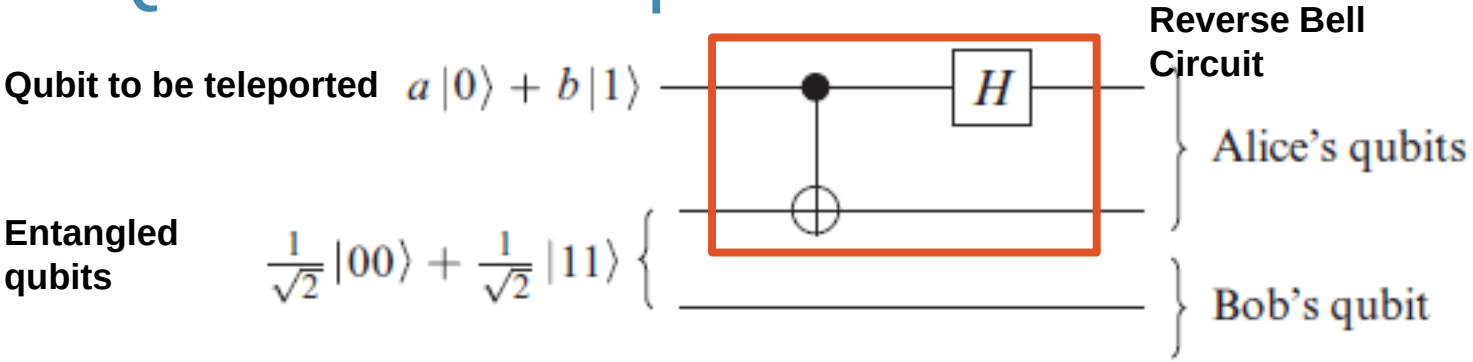
Alice's qubits

Bob's qubit

$$\frac{a}{2}|0\rangle\otimes|0\rangle\otimes|0\rangle + \frac{a}{2}|1\rangle\otimes|0\rangle\otimes|0\rangle + \frac{a}{2}|0\rangle\otimes|1\rangle\otimes|1\rangle$$
$$+\frac{a}{2}|1\rangle\otimes|1\rangle\otimes|1\rangle + \frac{b}{2}|0\rangle\otimes|1\rangle\otimes|0\rangle - \frac{b}{2}|1\rangle\otimes|1\rangle\otimes|0\rangle$$
$$+\frac{b}{2}|0\rangle\otimes|0\rangle\otimes|1\rangle - \frac{b}{2}|1\rangle\otimes|0\rangle\otimes|1\rangle$$

- **After H gate?**
- **Rearrange**

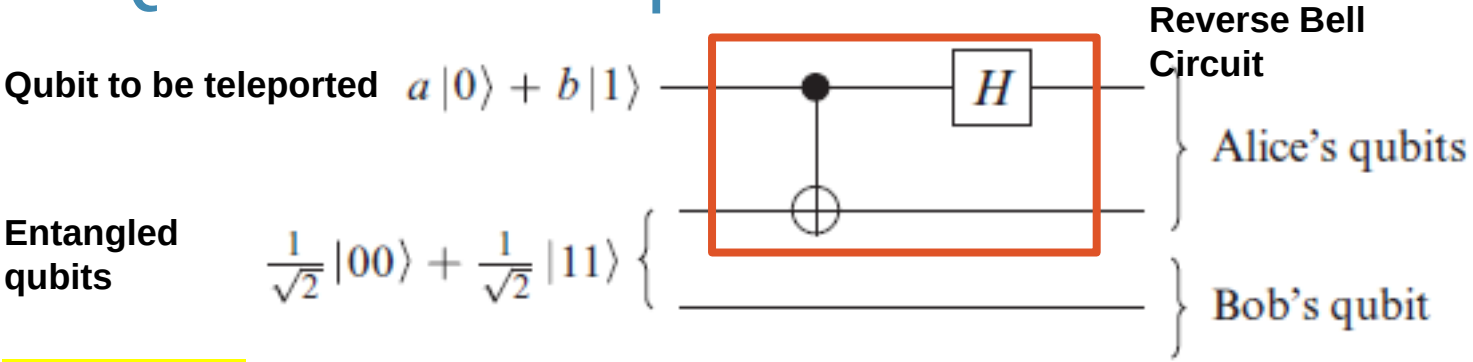
Quantum Teleportation – Full Circuit



- **What will happen now if Alice measures her two qubits?**

$$\frac{1}{2}|00\rangle\otimes(a|0\rangle+b|1\rangle)+\frac{1}{2}|01\rangle\otimes(a|1\rangle+b|0\rangle)$$
$$+\frac{1}{2}|10\rangle\otimes(a|0\rangle-b|1\rangle)+\frac{1}{2}|11\rangle\otimes(a|1\rangle-b|0\rangle)$$

Quantum Teleportation – Full Circuit

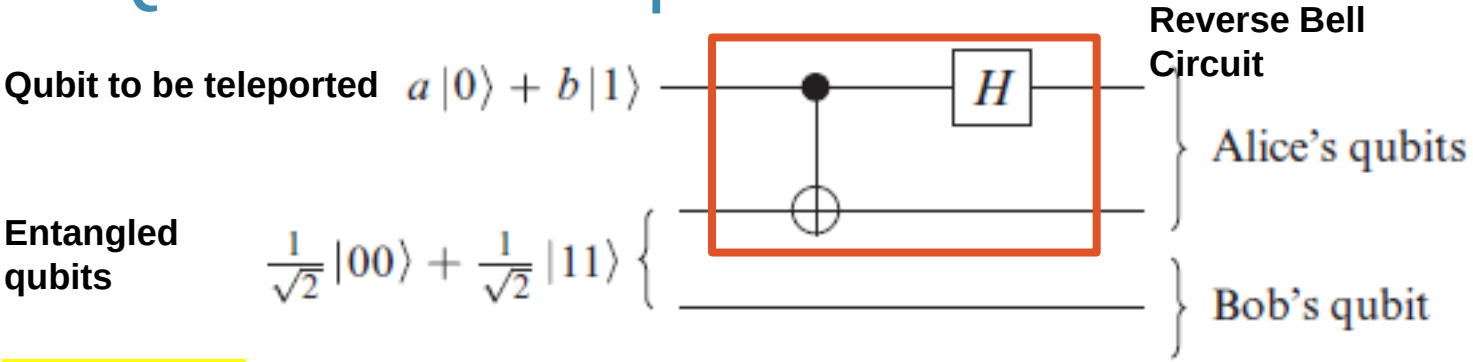


- **What will happen now if Alice measures her two qubits?**

$$\frac{1}{2}|00\rangle\otimes(a|0\rangle+b|1\rangle)+\frac{1}{2}|01\rangle\otimes(a|1\rangle+b|0\rangle)$$
$$+\frac{1}{2}|10\rangle\otimes(a|0\rangle-b|1\rangle)+\frac{1}{2}|11\rangle\otimes(a|1\rangle-b|0\rangle)$$

- **Will get 00, 01, 10, 11 with prob = 0.25**

Quantum Teleportation – Full Circuit

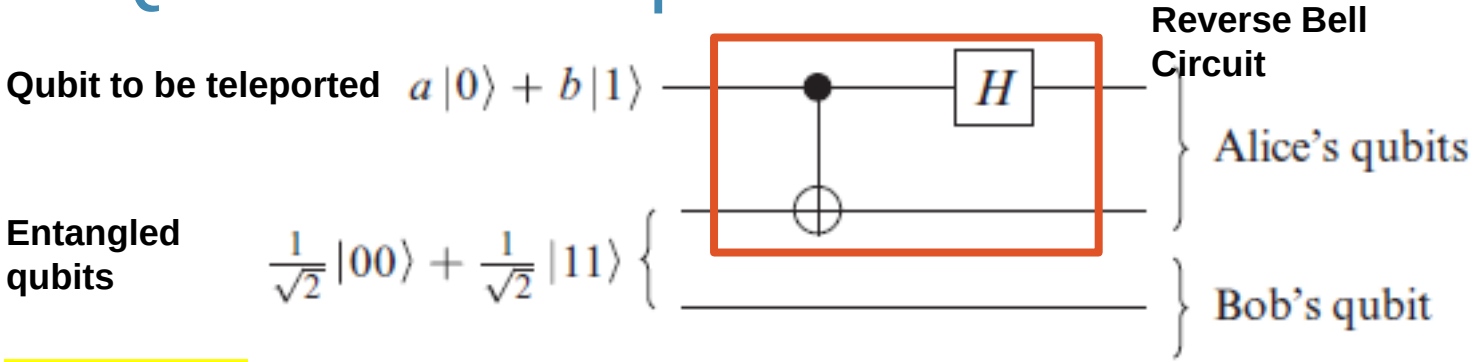


- **What will happen now if Alice measures her two qubits?**

$$\frac{1}{2}|00\rangle\otimes(a|0\rangle+b|1\rangle)+\frac{1}{2}|01\rangle\otimes(a|1\rangle+b|0\rangle)$$
$$+\frac{1}{2}|10\rangle\otimes(a|0\rangle-b|1\rangle)+\frac{1}{2}|11\rangle\otimes(a|1\rangle-b|0\rangle)$$

- **If she gets 00, Bob's qubit will jump to**

Quantum Teleportation – Full Circuit

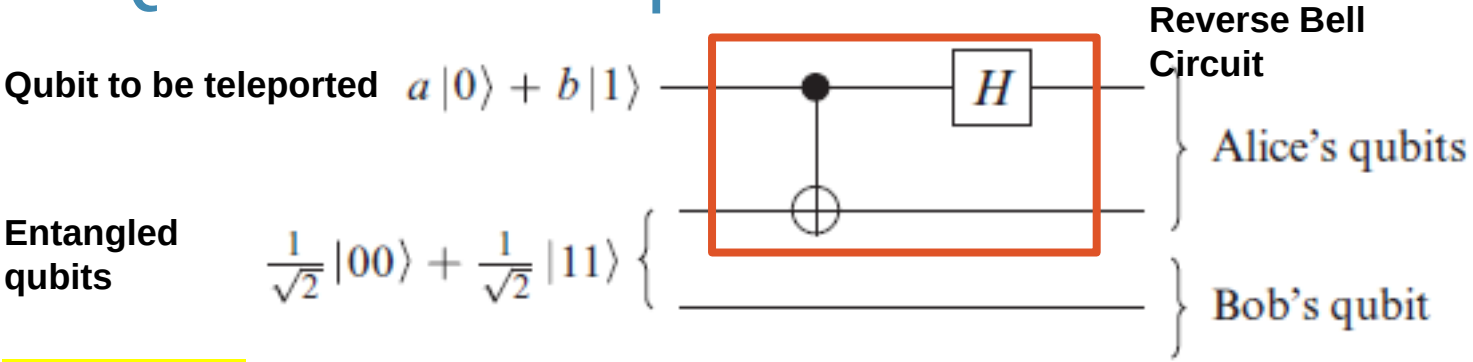


- **What will happen now if Alice measures her two qubits?**

$$\frac{1}{2}|00\rangle\otimes(a|0\rangle+b|1\rangle)+\frac{1}{2}|01\rangle\otimes(a|1\rangle+b|0\rangle)$$
$$+\frac{1}{2}|10\rangle\otimes(a|0\rangle-b|1\rangle)+\frac{1}{2}|11\rangle\otimes(a|1\rangle-b|0\rangle)$$

- **If she gets 01, Bob's qubit will jump to**

Quantum Teleportation – Full Circuit

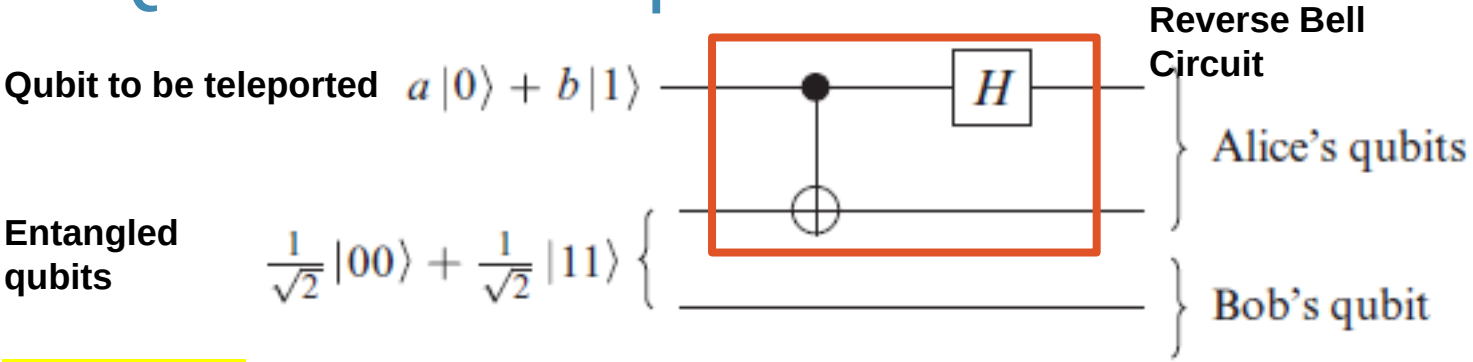


- **What will happen now if Alice measures her two qubits?**

$$\frac{1}{2}|00\rangle\otimes(a|0\rangle+b|1\rangle)+\frac{1}{2}|01\rangle\otimes(a|1\rangle+b|0\rangle)$$
$$+\frac{1}{2}|10\rangle\otimes(a|0\rangle-b|1\rangle)+\frac{1}{2}|11\rangle\otimes(a|1\rangle-b|0\rangle)$$

- **If she gets 10, Bob's qubit will jump to**

Quantum Teleportation – Full Circuit



- **What will happen now if Alice measures her two qubits?**

$$\frac{1}{2}|00\rangle\otimes(a|0\rangle+b|1\rangle)+\frac{1}{2}|01\rangle\otimes(a|1\rangle+b|0\rangle)$$
$$+\frac{1}{2}|10\rangle\otimes(a|0\rangle-b|1\rangle)+\frac{1}{2}|11\rangle\otimes(a|1\rangle-b|0\rangle)$$

- **If she gets 11, Bob's qubit will jump to**

Quantum Teleportation – Full Circuit

Alice after measurement	00	01	10	11
Bob's qubit after Alice measures	$a\|0\rangle + b\|1\rangle$	$a\|1\rangle + b\|0\rangle$	$a\|0\rangle - b\|1\rangle$	$a\|1\rangle - b\|0\rangle$

- **We want** $a|0\rangle + b|1\rangle$ **in all cases?**

Quantum Teleportation – Full Circuit

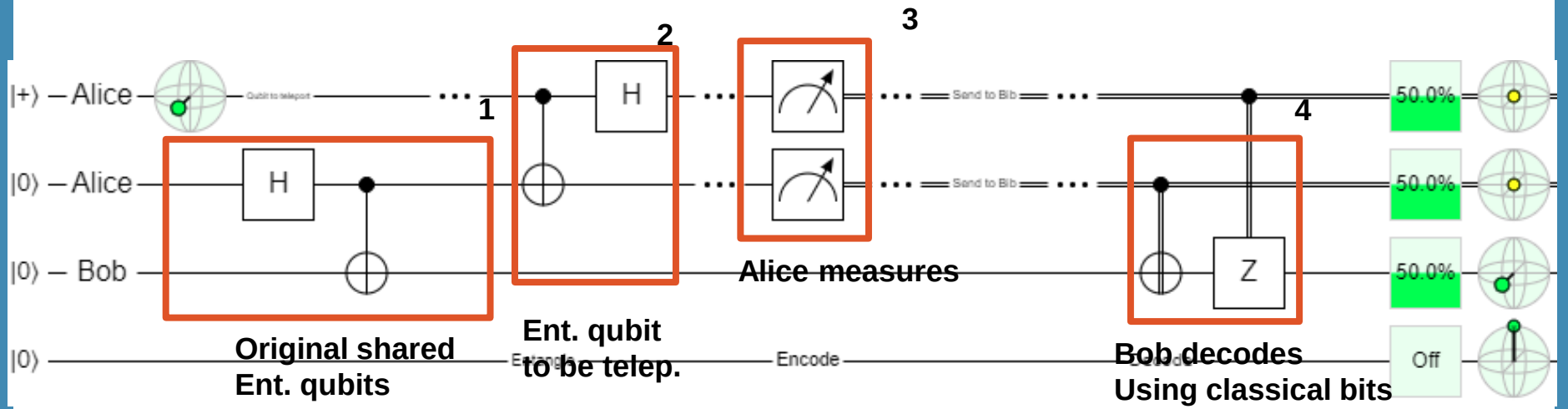
Alice after measurement	00	01	10	11
Bob's qubit after Alice measures	$a\|0\rangle + b\|1\rangle$	$a\|1\rangle + b\|0\rangle$	$a\|0\rangle - b\|1\rangle$	$a\|1\rangle - b\|0\rangle$

- **Alice sends her classical bits to Bob**
  - **Using any means (prevents superluminal communication)**
- **Bob gets 00 → Do nothing**
- **Bob gets 01 → Apply X gate**
- **Bob gets 10 → Apply Z gate**
- **Bob gets 11 → Apply Y gate**

Quantum Teleportation - Notes

- There is only one qubit in state $a_0|0> + a_1|1>$ at any point
  - Initially, Alice has it
  - At the end Bob has it
  - The no cloning theorem tells us
    - We can't copy, so only one of them can have it at a time
- Even though the state of Bob jumps immediately when Alice makes her measurement
  - Bob has to wait to know which one of the four states he has
  - By sending the two classical bits
  - Prevents superluminal communication

Quantum Teleportation - Demo



Agenda

- Introduction
- Background
- QC Algorithms Examples
- **QC for Positioning and Navigation**
- Conclusions and what’s next

QC for Positioning and Navigation

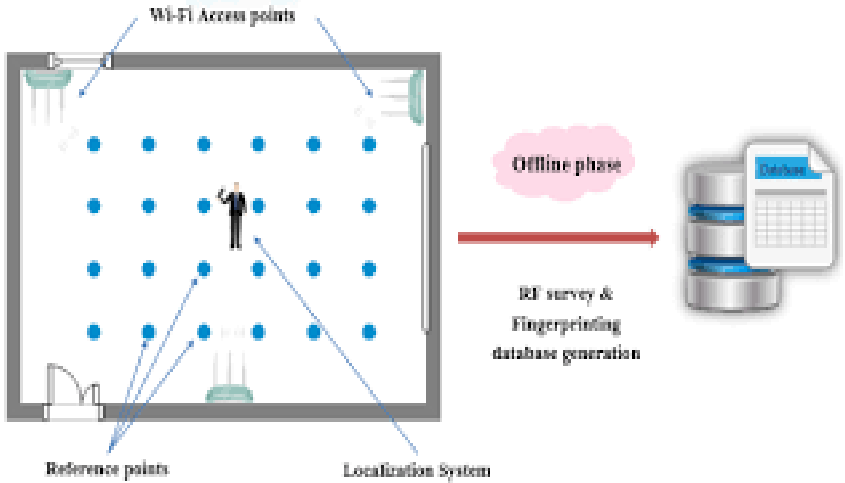
- General purpose QC
  - Speeding up processing
  - Exponential speedups

- Quantum Annealing

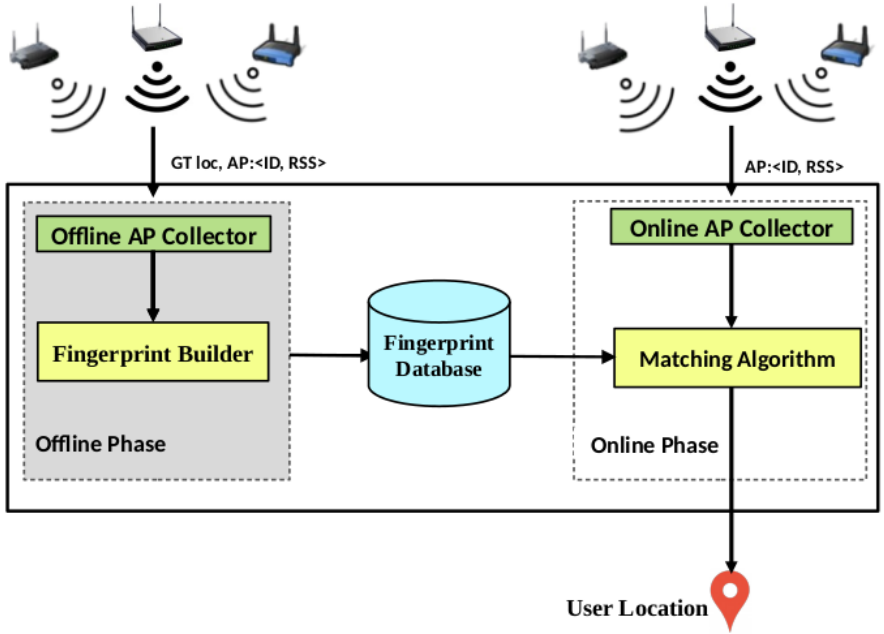
- Other Applications for Positioning and Navigation

Location Tracking Systems

- GPS doesn’t work indoors
- Propagation models do not provide good accuracy
  - Especially indoors
- Fingerprinting systems can provide high accuracy
  - Both indoors and outdoors

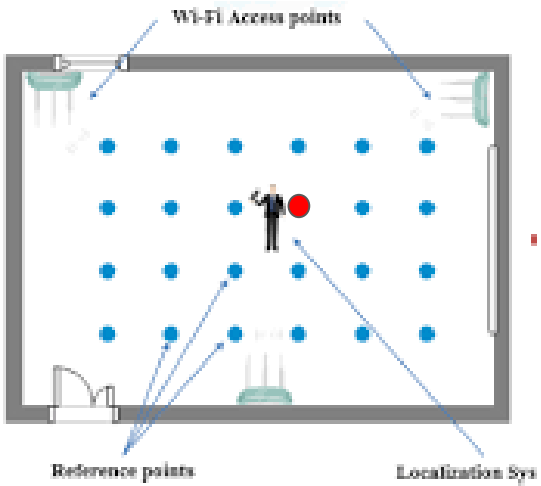


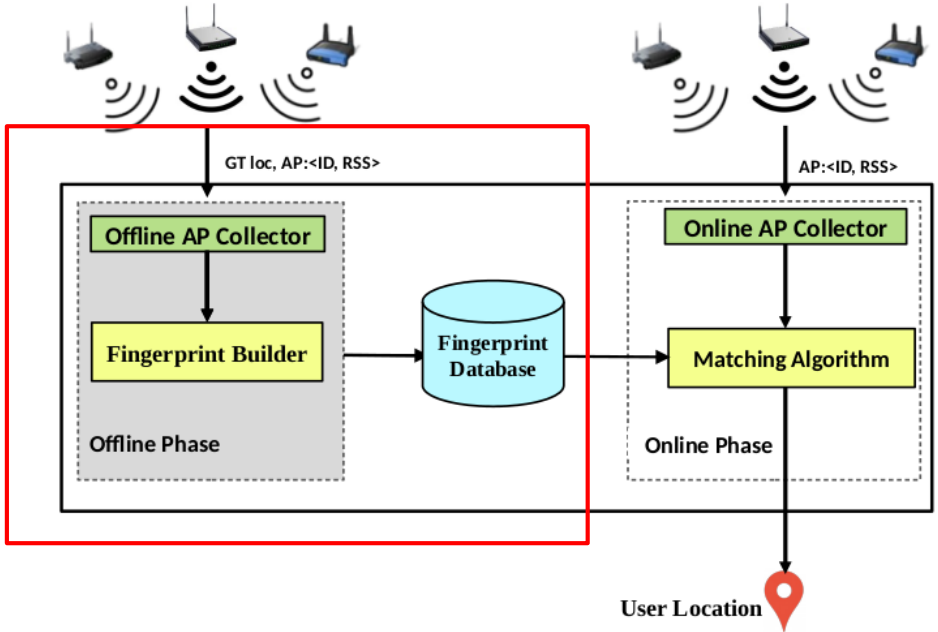
Fingerprinting Techniques - How They Work



Fingerprinting Techniques - Offline phase

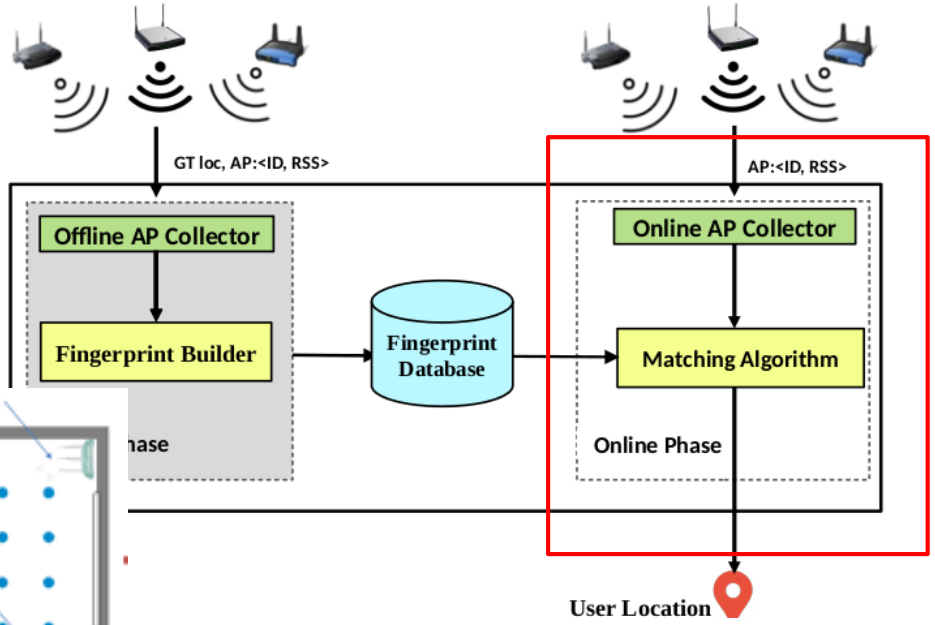
- Collect RSS training vectors from APs at known GT locations (fingerprint)

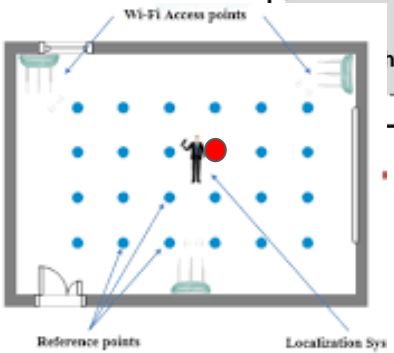




Fingerprinting Techniques - Online Phase

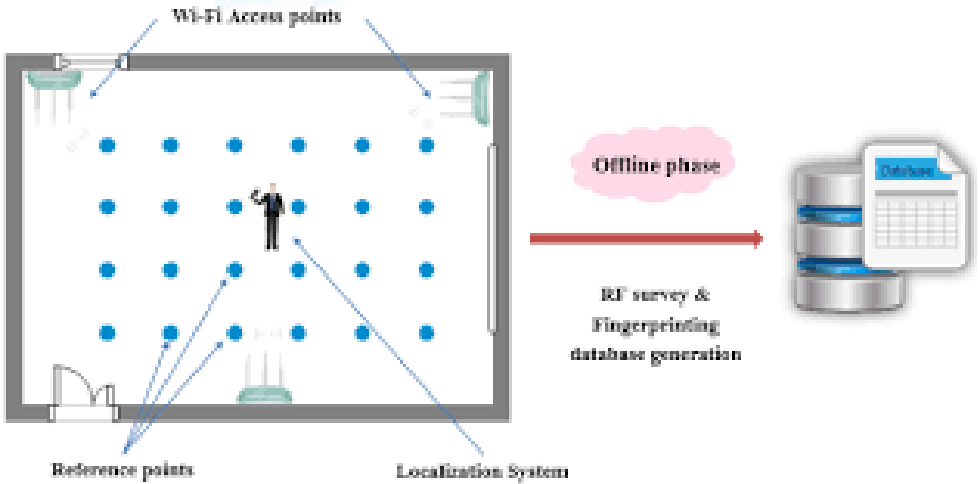
- Scan RSS from APs at unknown location
- Match the online user RSS vector with the fingerprint at the different locations.
- The closest location in the RSS space is reported as the estimated location





Fingerprinting-based Localization - Challenges

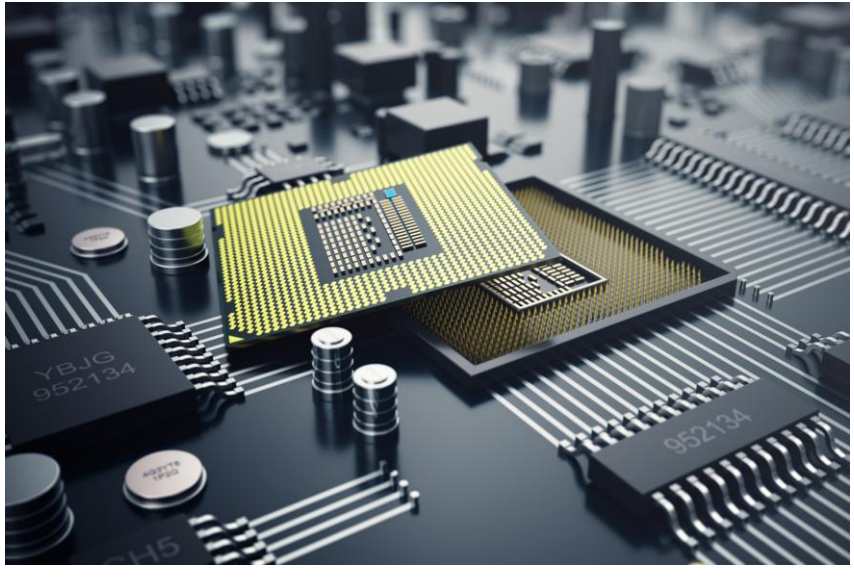
- Does not **scale** to a worldwide scale
  - The data storage **space** overhead.
  - The matching **processing** overhead.



Our Contribution

- Scale
  - Quantum Computing
    - **Exponential** saving in Space and Time

- Evaluated in **three real testbeds**
  - Indoor and Outdoor
  - WiFi and Cellular
  - On **Real Quantum Computers**

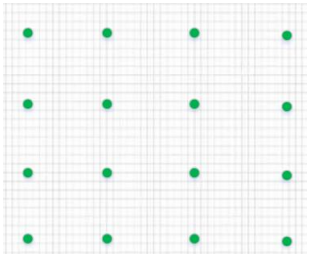


Quantum Localization

< Online RSS >

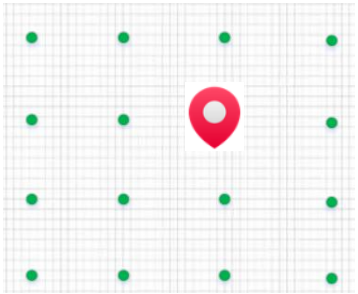


[< Offline RSS >]



Quantum State Preparation

Quantum Fingerprint Matching

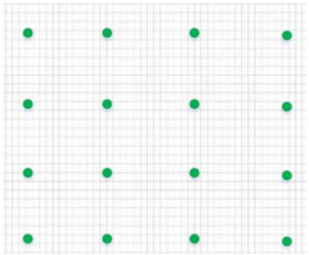


Quantum Localization

< Online RSS >

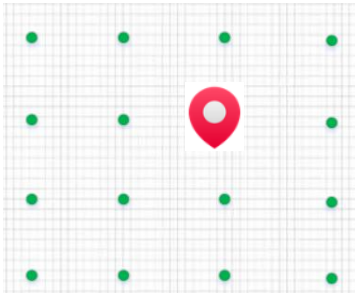


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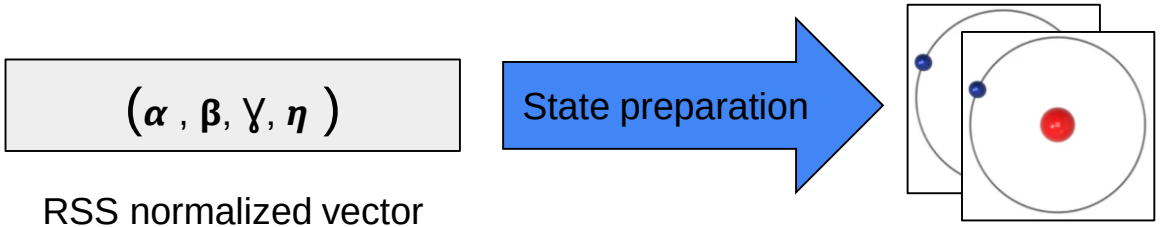
Quantum State Preparation

Quantum Fingerprint Matching



Quantum Localization - State Preparation

Encode the relative RSS vectors, coming from APs in the environment, in quantum registers



RSS normalized vector

Quantum register state α |00> + β |01> + ɣ |10> + η |11>

Exponential saving!

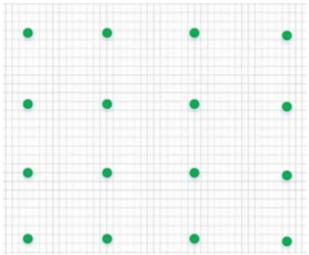
Using **amplitude encoding**

Quantum Localization

< Online RSS >

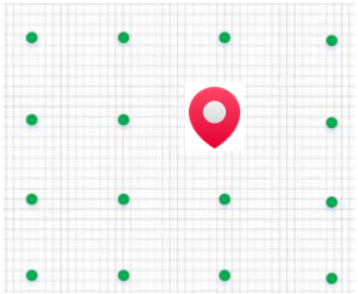


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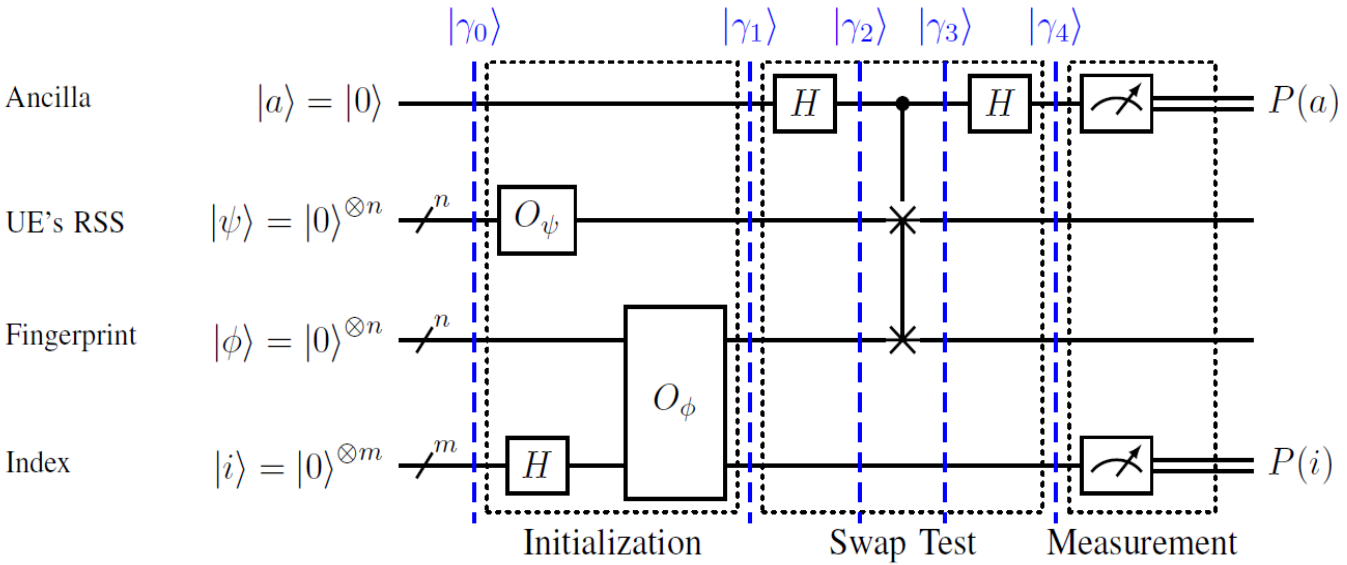


Quantum State Preparation

Quantum Fingerprint Matching

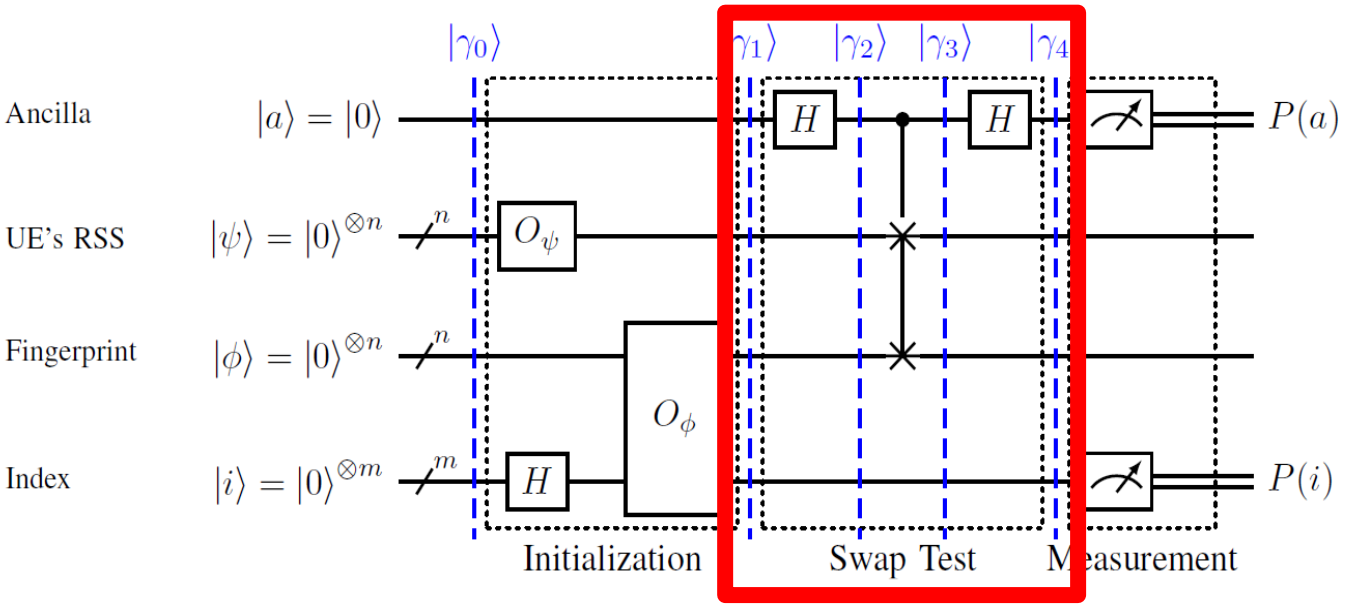


Quantum fingerprint matching circuit → O(log(MN))



Full Quantum Algorithm from our IEEE JSAC paper

Quantum fingerprint matching circuit → O(log(MN))



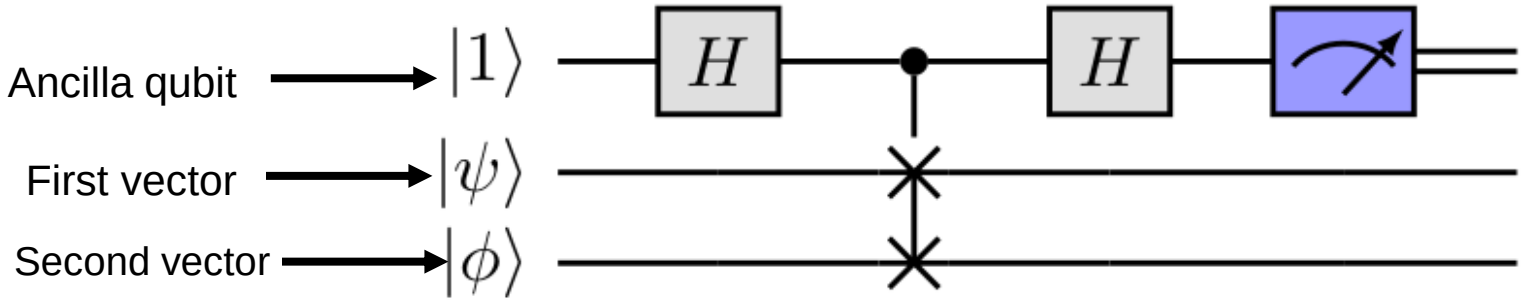
Full Quantum Algorithm from our IEEE JSAC paper

Quantum Cosine Similarity

Cosine Similarity Matching Circuit

The target of the circuit is to calculate the cosine similarity between two vectors

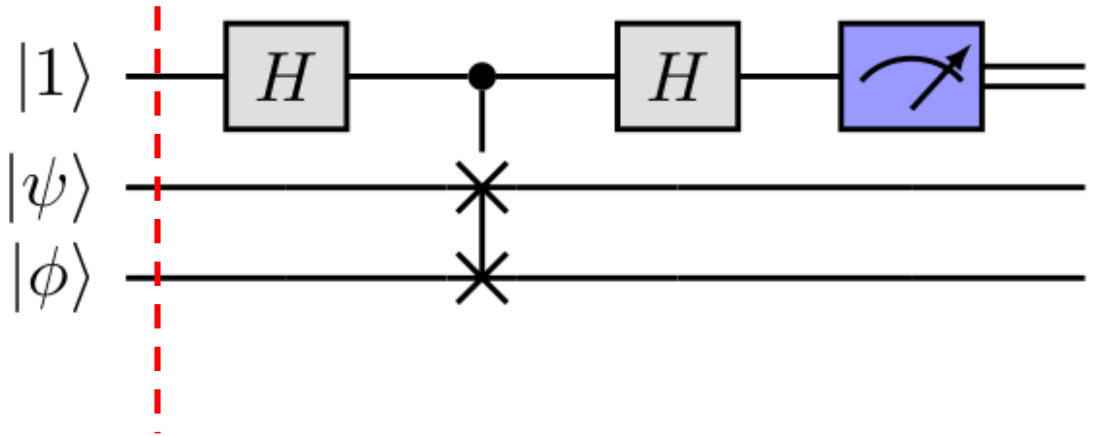
$$\mathrm{sim}(|\psi\rangle, |\phi\rangle) = \cos^2(\psi, \phi) = |\langle\phi|\psi\rangle|^2$$



Cosine Similarity Matching Circuit

The input to the circuit is

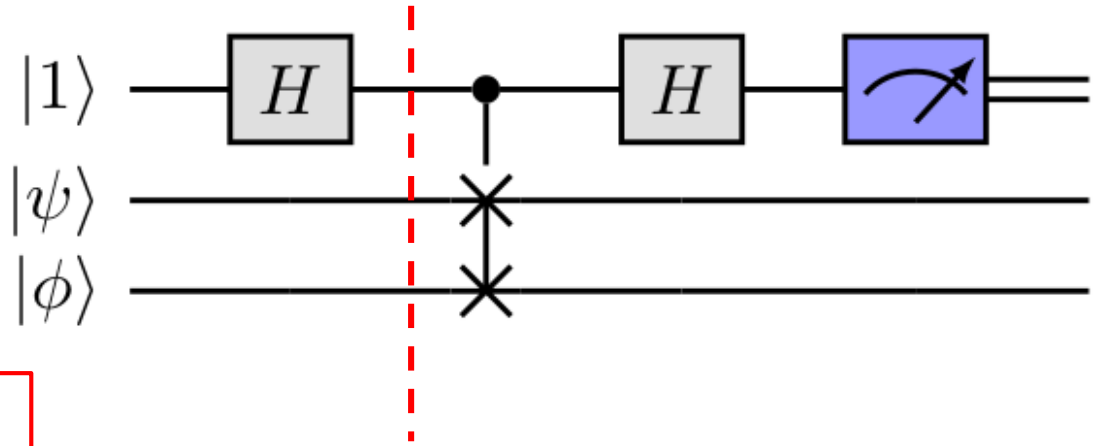
$|1\rangle |\Psi\rangle |\phi\rangle$



Cosine Similarity Matching Circuit

After applying the H-gate to the ancilla qubit (|1>) we obtain the superposition state:

|1> |Ψ> |ϕ>



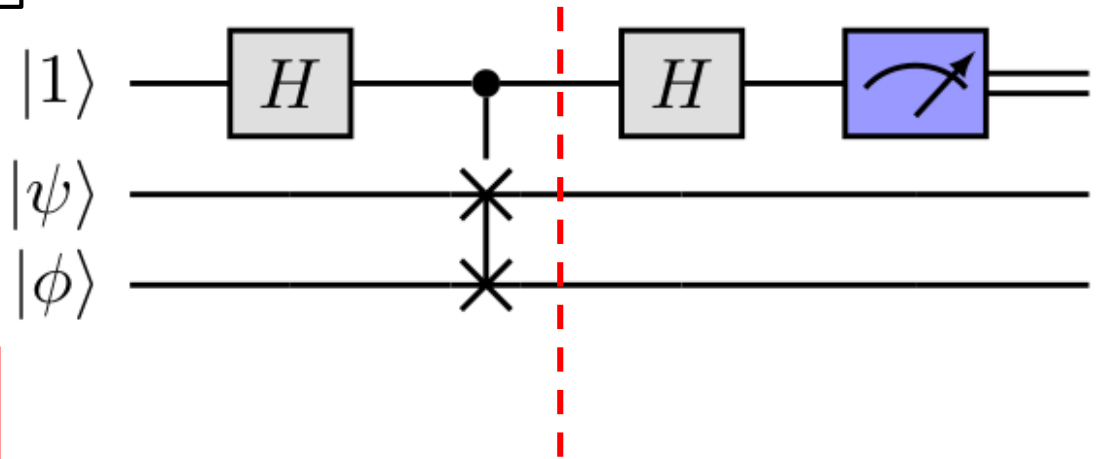
$$\frac{1}{\sqrt{2}}(|0\rangle\,|\psi\rangle\,|\phi\rangle - |1\rangle\,|\psi\rangle\,|\phi\rangle)$$

Cosine Similarity Matching Circuit

After applying the CSWAP-gate, which exchanges the state when the ancilla is |1>, we obtain the superposition state:

$$\frac{1}{\sqrt{2}}(|0\rangle\,|\psi\rangle\,|\phi\rangle - |1\rangle\,|\psi\rangle\,|\phi\rangle)$$

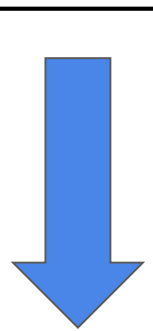
$$\frac{1}{\sqrt{2}}(|0\rangle\,|\psi\rangle\,|\phi\rangle - |1\rangle\,|\phi\rangle\,|\psi\rangle)$$

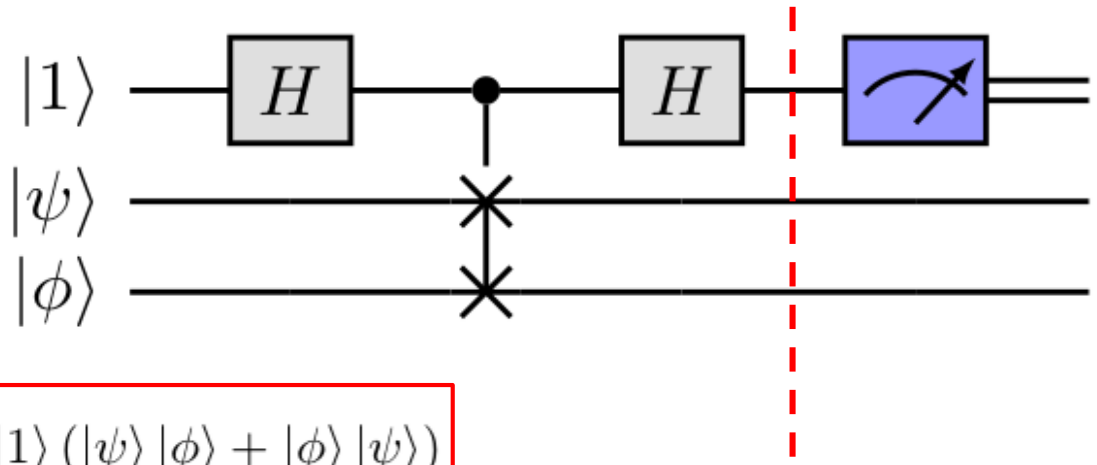


Cosine Similarity Matching Circuit

The state after applying the second H-gate to the ancilla qubit:

$$\frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle|\phi\rangle - |1\rangle|\phi\rangle|\psi\rangle)$$





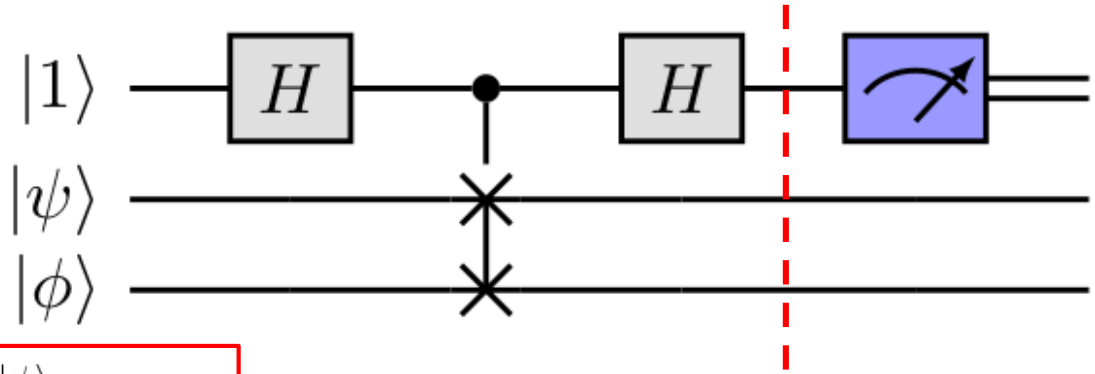
$$\frac{1}{2}|0\rangle(|\psi\rangle|\phi\rangle - |\phi\rangle|\psi\rangle) + \frac{1}{2}|1\rangle(|\psi\rangle|\phi\rangle + |\phi\rangle|\psi\rangle)$$

Cosine Similarity Matching Circuit

Finally, by normalizing the state, the circuit will be in the following joint state,

$$\frac{1}{2}|0\rangle\left(|\psi\rangle|\phi\rangle - |\phi\rangle|\psi\rangle\right) + \frac{1}{2}|1\rangle\left(|\psi\rangle|\phi\rangle + |\phi\rangle|\psi\rangle\right)$$



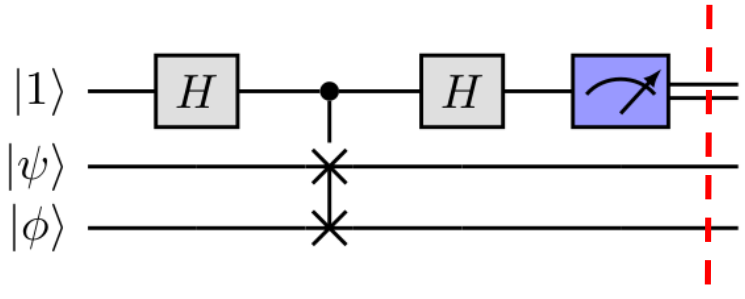


$$\frac{1}{2}\left(\sqrt{2-2|\langle\psi|\phi\rangle|^2}\,|0\rangle\,\frac{|\psi\rangle|\phi\rangle - |\phi\rangle|\psi\rangle}{\sqrt{2-2|\langle\psi|\phi\rangle|^2}} + \sqrt{2+2|\langle\psi|\phi\rangle|^2}\,|1\rangle\,\frac{|\psi\rangle|\phi\rangle + |\phi\rangle|\psi\rangle}{\sqrt{2+2|\langle\psi|\phi\rangle|^2}}\right)$$

Cosine Similarity Matching Circuit

The target of the circuit is to calculate the cosine similarity between online RSS and offline ones from the fingerprint.

$$\mathrm{sim}(|\psi\rangle, |\phi\rangle) = \cos^2(\psi, \phi) = |\langle\phi|\psi\rangle|^2$$



- After applying the different gates, the

$$\frac{1}{2}\left(\sqrt{2-2|\langle\psi|\phi\rangle|^2}\,|0\rangle\,\frac{|\psi\rangle|\phi\rangle - |\phi\rangle|\psi\rangle}{\sqrt{2-2|\langle\psi|\phi\rangle|^2}} + \sqrt{2+2|\langle\psi|\phi\rangle|^2}\,|1\rangle\,\frac{|\psi\rangle|\phi\rangle + |\phi\rangle|\psi\rangle}{\sqrt{2+2|\langle\psi|\phi\rangle|^2}}\right)$$

- After running the circuit K times and measuring its output, the similarity will be, $2\times\#|1\rangle/K-1.$

Classical vs Quantum Complexity Analysis

Space complexity:

Classical	Quantum

- **M is the number of fingerprint locations.**
- **N is the number if RPs in the environment.**

Classical vs Quantum Complexity Analysis

Space complexity:

Classical	Quantum
o(N)	

- **N is the vector length.**

Classical vs Quantum Complexity Analysis

Space complexity:

Classical		Quantum
o(N)	→	**o(logN)**

- **N is the vector length.**

Classical vs Quantum Complexity Analysis

Space complexity:

Classical	Quantum
o(N)	**o(logN)**

Time complexity:

Classical	Quantum

- **N is the vector length.**

Classical vs Quantum Complexity Analysis

Space complexity:

Classical	Quantum
o(N)	**o(logN)**



Time complexity:

Classical	Quantum
o(N)	

- **N is the vector length.**

Classical vs Quantum Complexity Analysis

Space complexity:

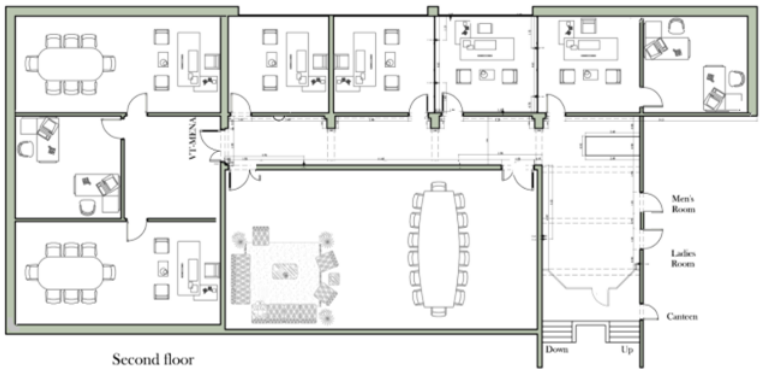
Classical o(N) → Quantum **o(logN)**

Time complexity:

Classical o(N) → Quantum **o(logN)**

- **N is the vector length.**

Results - Testbeds Summary



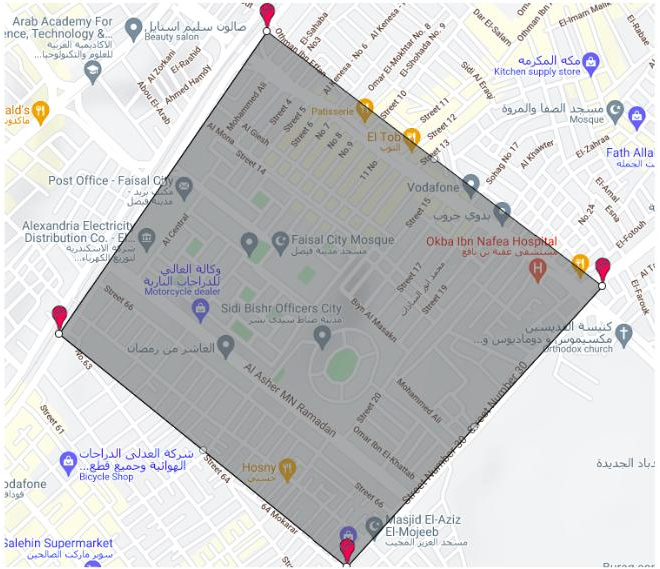


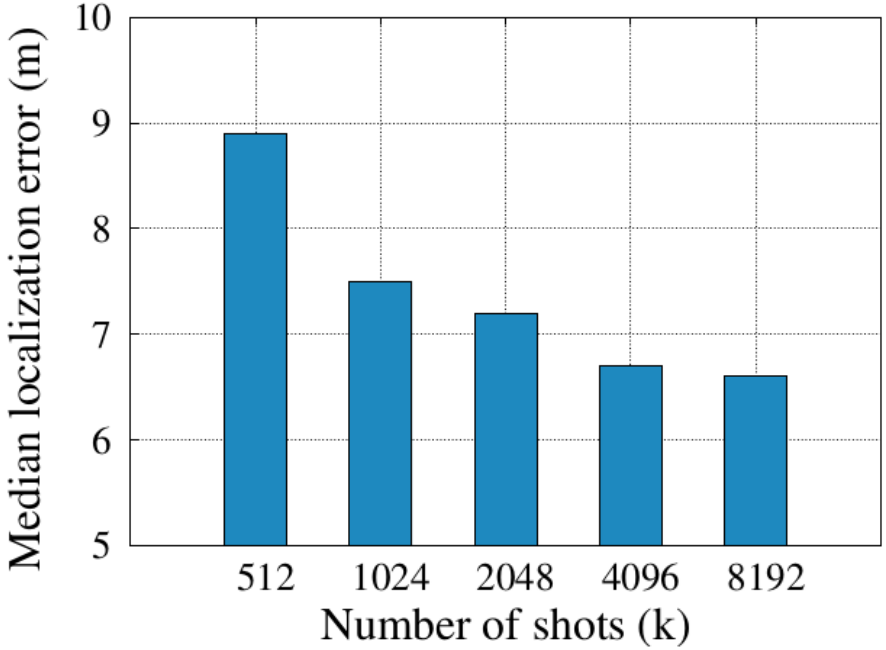
Fig. 10: The outdoor testbed.

Testbeds	Deployment	Area	Tectnology	Quantum machine	Reference points	Used qubits
Testbed 1	Indoor	89ft × 56ft	WiFi	*ibmq_santiago* (5 qubits)	4 (APs)	5
Testbed 2	Indoor	89ft × 56ft	WiFi	*qasm_simulator* (28 qubits)	16 (APs)	9
Testbed 3	Outdoor	0.2Km^2	Cellular	*qasm_simulator* (28 qubits)	21 (Cell towers)	11

Results - Shots

- Effect of number of shots on localization accuracy.

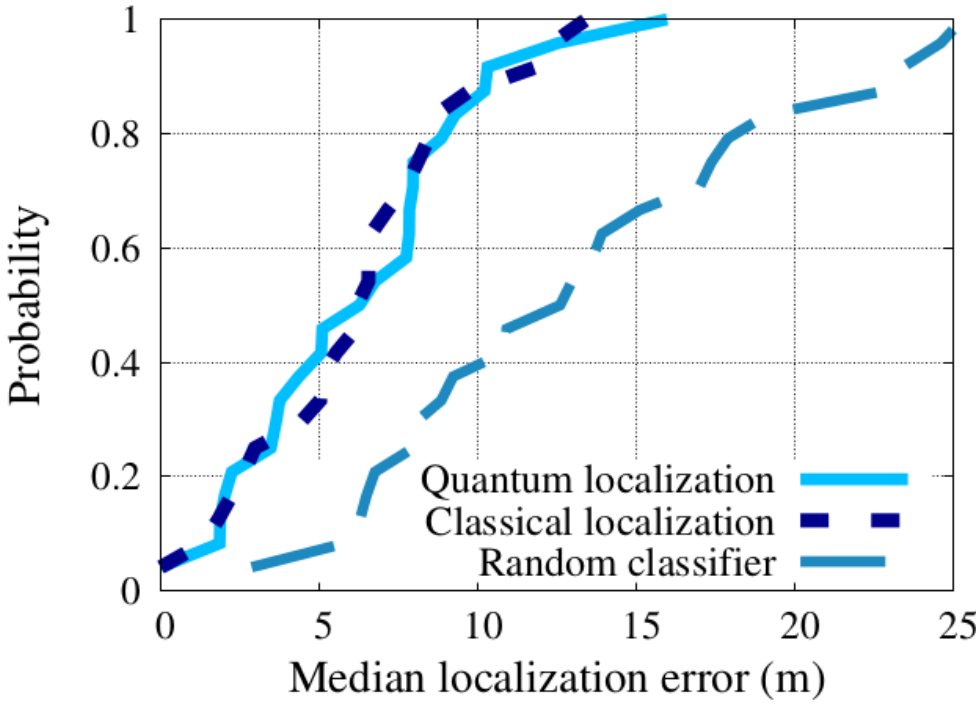
- Increasing the number of iterations leads to a better localization accuracy till it saturates around 4096 shots.



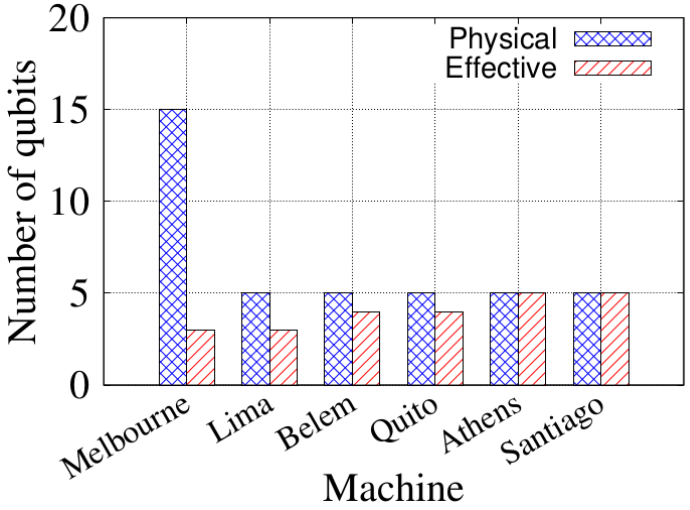
Results - CDFs

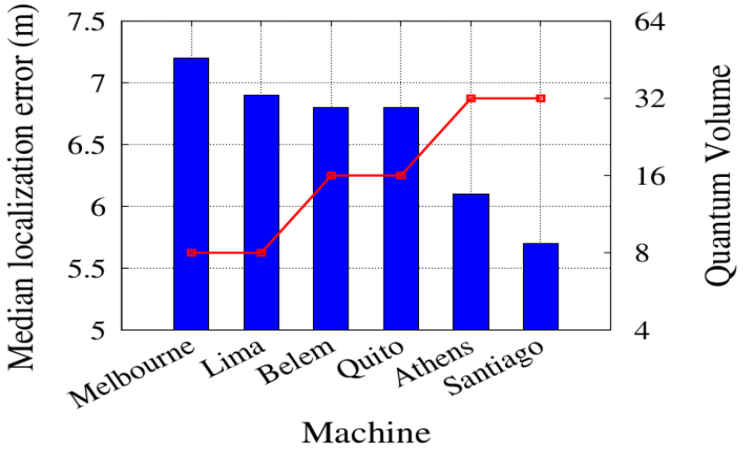
- CDF of localization error.

- Quantum and classical fingerprinting localization systems have the same performance.



Results - Effect of Machine Error Profile on Localization

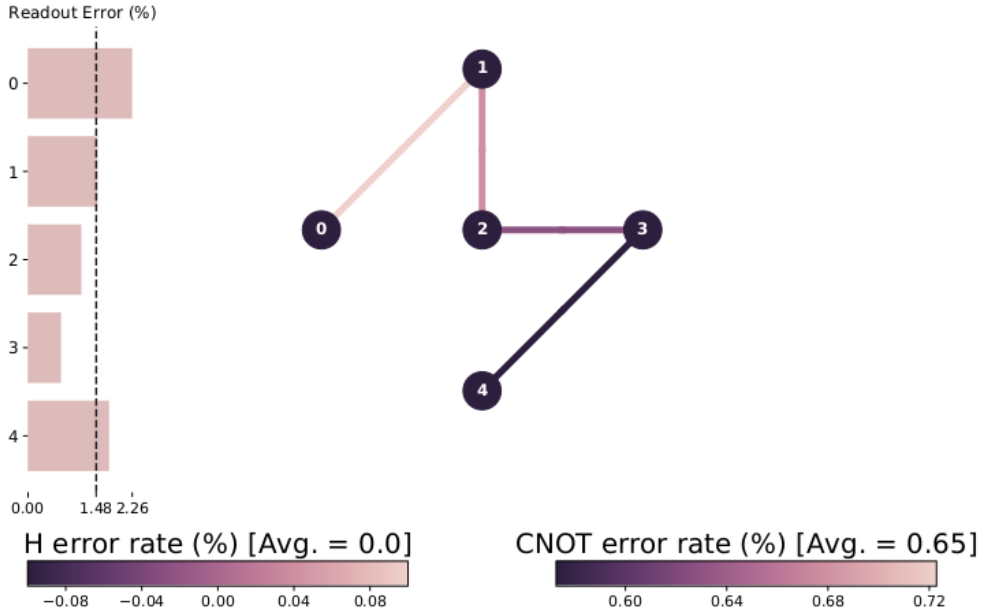




Machine	T	p(0|1)	p(1|0)	Readout error	H gate error	CNOT gate error	#qubits	QV
ibmq_santiago	146.1μs	0.01844	0.00888	0.0137	1.887e-4	6.740e-3	5	32
ibmq_athens	77.58μs	0.03904	0.0128	0.0259	2.345e-3	2.107e-2	5	32
ibmq_quito	86.25μs	0.04232	0.01024	0.0263	1.165e-3	1.107e-2	5	16
ibmq_belem	82.10μs	0.048	0.01004	0.0290	2.908e-4	1.003e-2	5	16
ibmq_lima	85.16μs	0.0422	0.00998	0.0261	6.266e-4	1.570e-2	5	8
ibmq_16_melbourne	56.49μs	0.048	0.022	0.0350	6.557e-3	4.308e-2	15	8

Results - IBM Q Santiago Machine

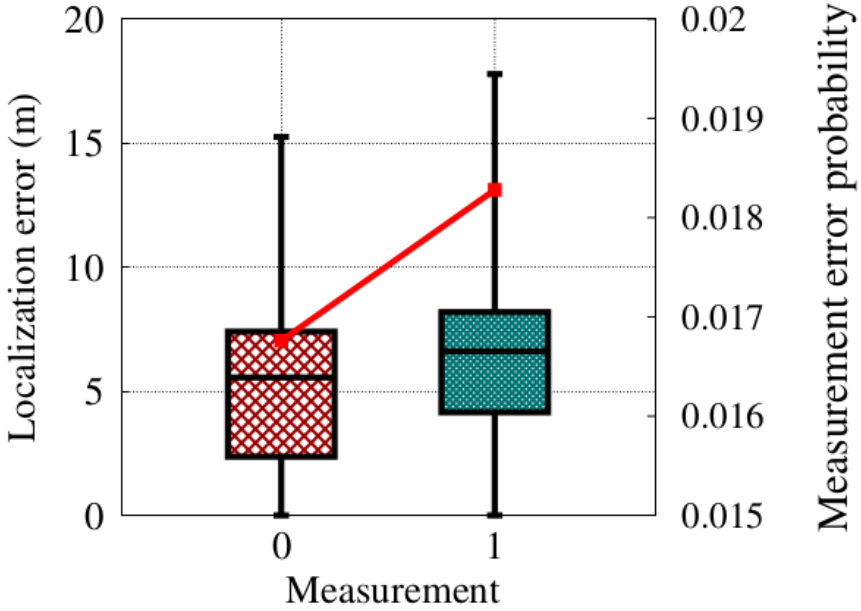
- 5 qubits
- Vertex color: single qubit gate error (e.g. H)
- Arcs color: Two-qubits gates error (e.g. CNOT)
- Vertical line is measurement error



Results - Measurement Probability

- Effect of measurement error probability on the localization accuracy

- Measuring the ancilla qubit to be in 0 state will lead to a lower error, since p(1|0) < p(0|1) for the IBM Santiago machine



QC for Positioning and Navigation

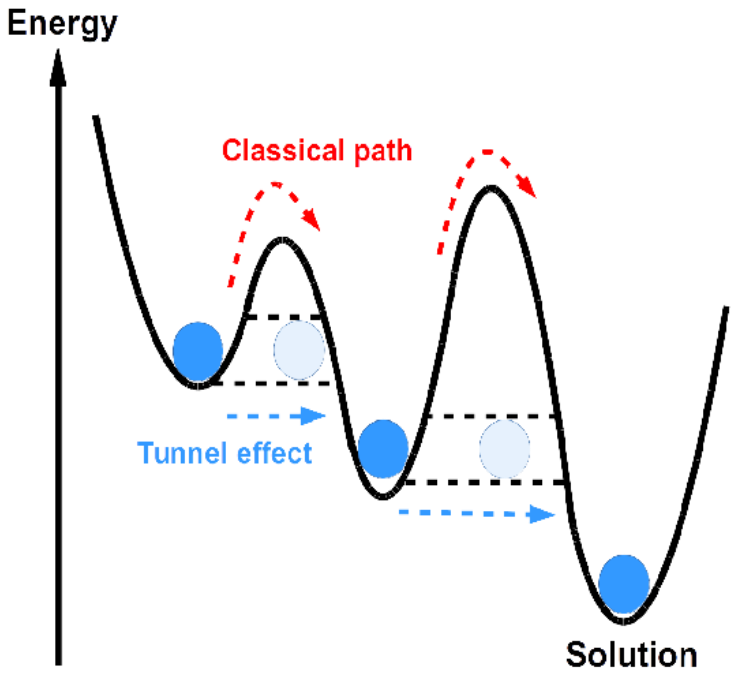
- General purpose QC
  - Speeding up processing
  - Exponential speedups
- **Quantum Annealing**
- Other Applications for Positioning and Navigation

Quantum Annealing

Quantum Annealing

→ Quantum Annealing finds the optimal solution of problems involving a large number of solutions

→ It takes advantage of quantum properties like quantum **superposition** and **tunneling** to obtain the optimal solution faster than the classical solver



Quantum Tunnelling

Quantum Annealing

- Most quantum annealers that can efficiently solve optimization problems as long as you can reduce the problem to QUBO problem

What is QUBO?

- Quadratic Unconstrained Binary Optimization

$$\text{Minimize } f(\mathbf{x}) = \sum_{i,j} Q_{ij} x_i x_j + \sum_i P_i x_i$$

- Where:
  → f(x) is the objective function
  → x is a vector that represents the binary variables
  → Qij are coefficients representing the quadratic terms
  → Pi are coefficients representing the linear terms

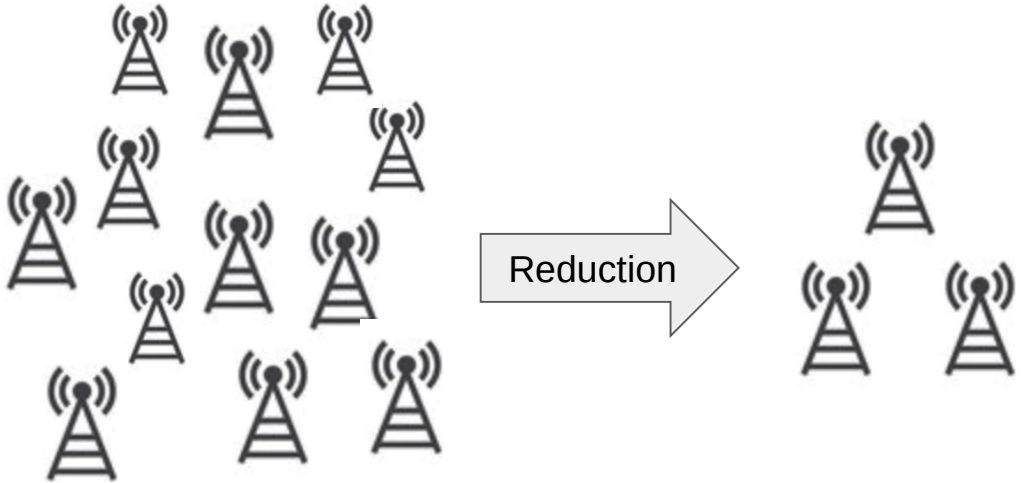
- Most quantum solvers need to convert the optimization problem into QUBO form

Example: APs Selection for Localization

- We introduce quantum APs selection algorithm for large scale localization.
  → Reduce storage space
  → Reduce running time overhead
  → Improve robustness

- Deployment on a real quantum machine (D-Wave)

- Deployment in a real environment



Our Problem

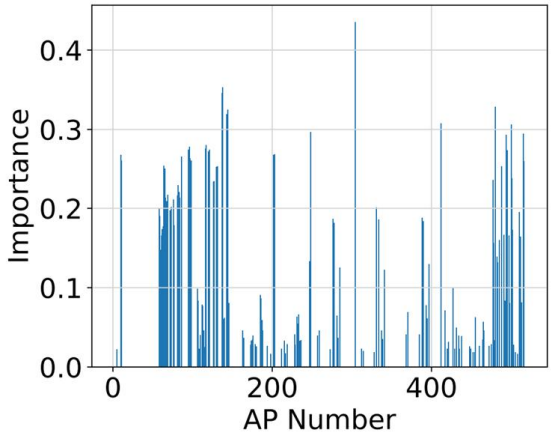
- We can formulate the AP selection problem in terms of QUBO, where the presence of a AP is a binary variable of value 1, and the absence of a feature is a variable equal to 0.

Basic idea - How to select APs?

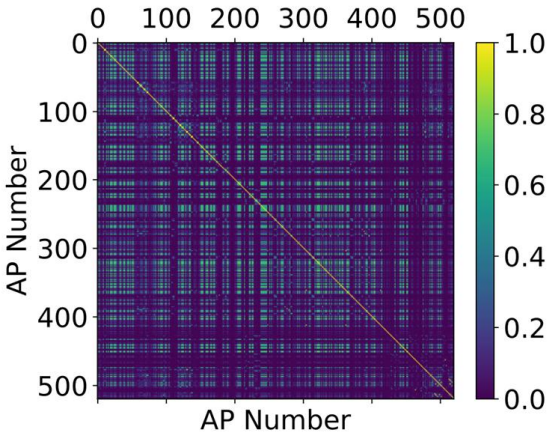
- Select APs that have

→ **Highest importance:**
The highest influence on estimating the location

→ **Lowest Redundancy:**
Tries to reduce the APs that depend on each others



(a) Importance



(b) Redundancy

Fig. 1: Importance and Redundancy in the full dataset. The importance reflects the influence of each AP on estimating the location. The Redundancy reflects APs inter-dependency.

Importance

- We define the importance of each AP i as the Cramer's V
  → measure of the signal strength from AP association with the output labels.

- Importance ranges from 0 to 1:
  → 0 no association between the AP RSS and output label
  → 1 indicates a strong association

$$I_i = \sqrt{\frac{\chi^2}{m \times \min(\text{b-1}, \text{f-1})}}$$

→ Where $\chi 2$ is the chi-square statistic
→ m is the number of fingerprint samples, f is the number of distinct labels
→ b is the number of possible RSS values.

Redundancy

- We define the redundancy between each pair of APs, i and j, as the strength of the absolute Pearson correlation between them.

$$R_{ij} = |\mathrm{Corr}(r_i, r_j)| \geq 0$$

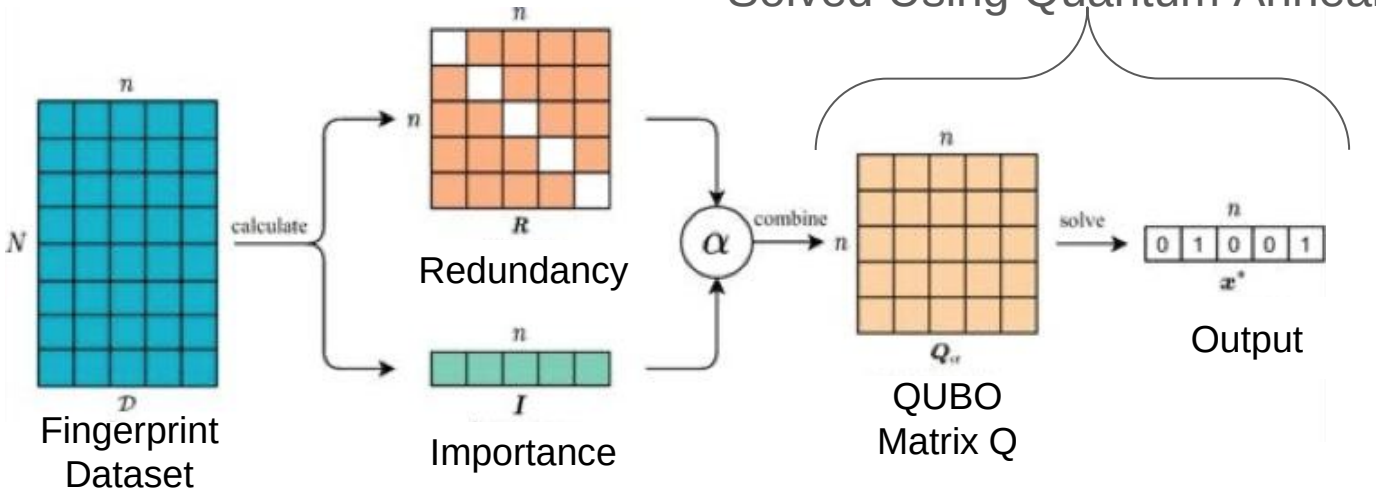
- Range of redundancy between 0 and 1:
  → 0 means totally independent
  → 1 means totally dependent

QUBO Formulation

Objective: $$\boldsymbol{x}^* = \underset{\boldsymbol{x}\in\{0,1\}^n}{\arg\min}\, Q(\boldsymbol{x},\alpha) = -\alpha \sum_{i=1}^{n} I_i x_i + (1-\alpha) \sum_{i,j=1}^{n} R_{ij} x_i x_j,$$

Where **x** is a binary variable indicating whether AP is selected or not and **α** is a balancing parameter that balances the importance and the redundancy.

Solved Using Quantum Annealing



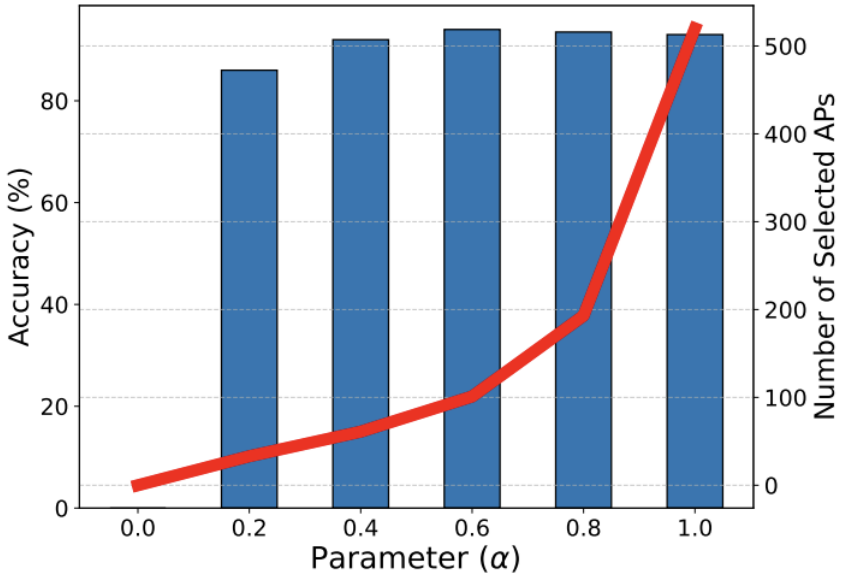
Parameter α

$$\boldsymbol{x}^* = \underset{\boldsymbol{x}\in\{0,1\}^n}{\arg\min}\, Q(\boldsymbol{x}, \alpha) = -\alpha \sum_{i=1}^{n} I_i x_i + (1-\alpha) \sum_{i,j=1}^{n} R_{ij} x_i x_j,$$

→ **α=0:** minimizes redundancy
No AP is selected

→ **α=1:** maximize the importance
Select all APs

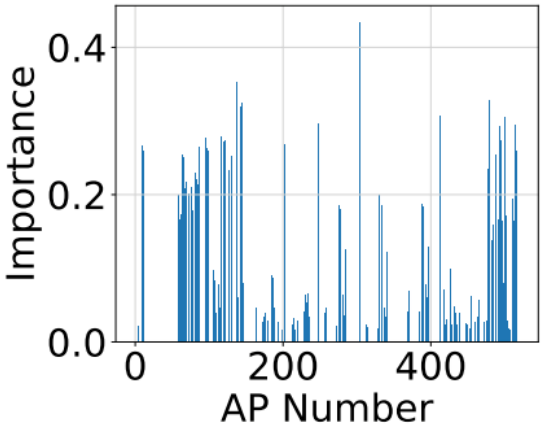
→ Parameter α can control the number of selected APs which in turn affects the accuracy



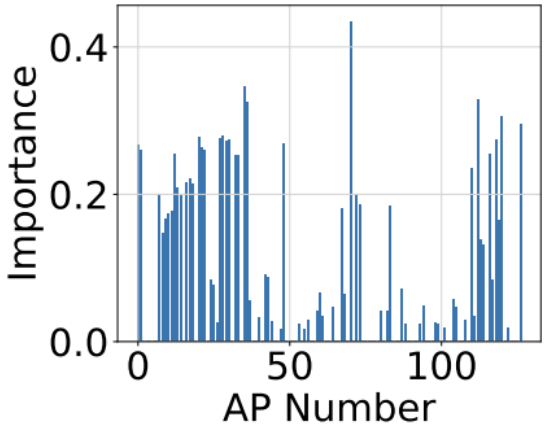
Machine & Environment Setup

- Real D-Wave hybrid solver with 5000 qubits
- Public dataset with 5 floors
- Area 110m2
- Number of APs is 520
- Number of samples 10k

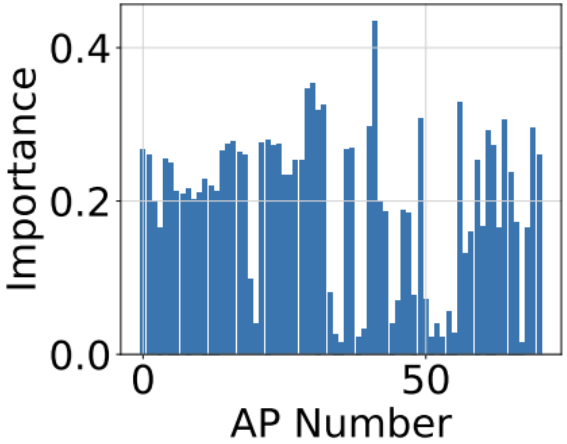
Importance



Complete Dataset

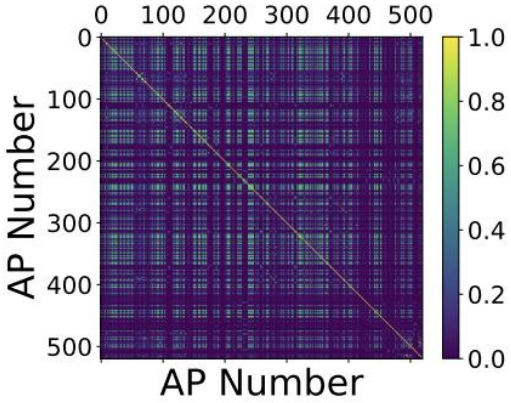


Simulated Annealing



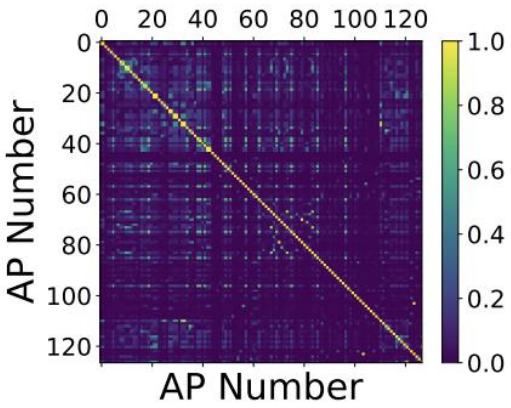
Quantum Annealing

Redundancy



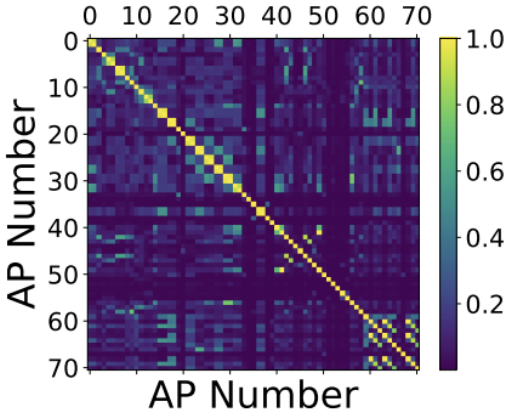
(b) Redundancy

Complete Dataset



(b) Redundancy

Simulated Annealing

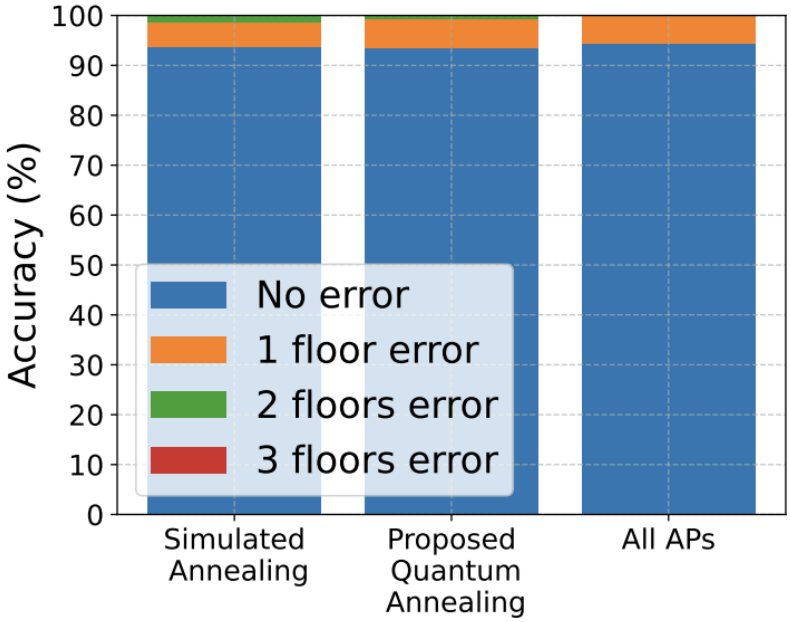


(b) Redundancy

Quantum Annealing

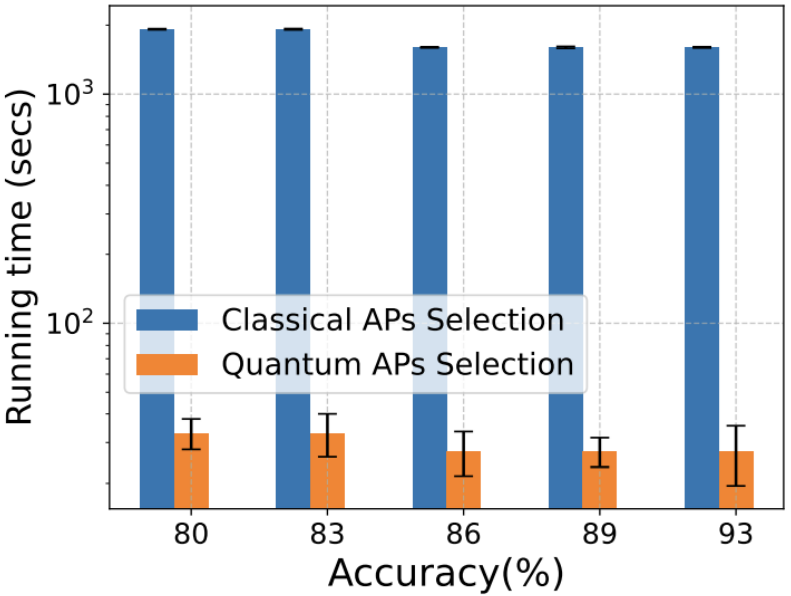
Accuracy

→ Floor localization accuracy using the classical simulated annealing algorithm and the quantum annealing algorithm.



Running time

→ Running time of classical and quantum APs selection algorithms.



QC for Positioning and Navigation

- General purpose QC
  - Speeding up processing
  - Exponential speedups
- Quantum Annealing
- **Other Applications for Positioning and Navigation**

Quantum Sensing

Quantum Sensing

- Quantum sensors are based on quantum systems
- Can provide measurements with a **sensitivity** that is **orders of magnitude** > **classical sensing**
- More mature than other quantum technologies
- Could lead to unobservables that can now be observed
  - New insights and data sets across many sectors of the economy
- Potential applications
  - Imaging and diagnostics, **navigation**, fault analysis, and underground measurements

Quantum Sensors

- Sense Physical properties:
  - Magnetic fields
  - Electric fields
  - Rotation
  - Acceleration
  - Temperature
  - Gravity
  - Time
  - Pressure
  - etc

Quantum Sensors

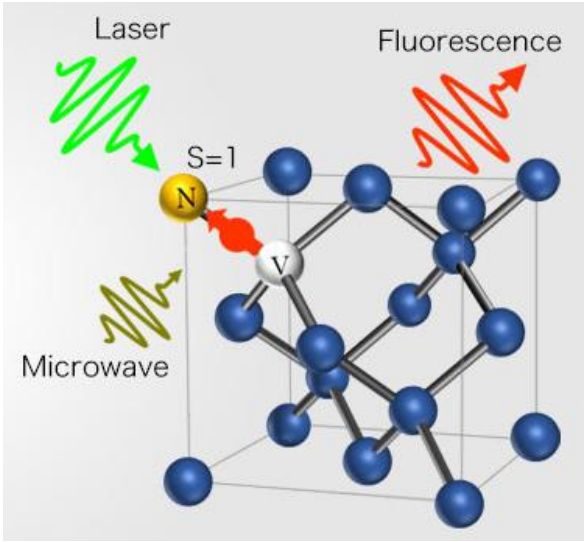
- Sense Physical properties:
  - **Magnetic fields**
  - Electric fields
  - **Rotation**
  - **Acceleration**
  - Temperature
  - **Gravity**
  - **Time**
  - **Pressure**
  - Etc
- Potential use for **positioning and navigation**

Quantum Sensors in Navigation

- Using it **today** in **Atomic Clocks** in satellites
- **Near term**
  - Quantum sensor-based navigation
    - In GPS-denied areas
      - Or jamming scenarios!
      - Udnerwater, undergroud, space
    - Deadreckoning based?
- Quantum sensors require calibration
  - Far less frequently
  - 1-2 orders of magnitude more sensitive than conventional technology

Quantum Sensors in Navigation

- Need to make quantum navigation sensors
  - Compact, robust, and low-cost
  - Field tests of prototype quantum navigation systems suggest that nitrogen vacancy (NV) diamond sensors and optically pumped magnetometers based on neutral atoms
    - May be ready for commercial use within the **next few years**



Other Areas

- Quantum Encryption
  - Secure Localization
  - Location verification
  - No-cloning theory
- Quantum Networking and Communication
  - Transport Qubits
  - No-cloning theory
  - Entanglement
- Quantum ML

Agenda

- Introduction
- Background
- QC Algorithms Examples
- QC for Positioning and Navigation
- **Conclusions and what’s next**

Conclusion

- QC is a promising new field.
- Requires a **new way** of thinking about designing algorithms.
- Quantum algorithms can provide **exponential** saving in the space and the running time of cosine similarity calculation.
  - And other algorithms

(c) Moustafa Youssef

Tools

- IBM Cloud Quantum Computer
  - Make your account today
- Quirk

References

- Our vision paper in SIGSpatial 2021 here.
  - https://dl.acm.org/doi/10.1145/3474717.3483958
  - A longer version here.
  - https://arxiv.org/abs/2106.11751
- Algorithm described in the presentation in details here
  - Best paper runner up.
  - https://ieeexplore.ieee.org/document/9843246/
- More detailed analysis of practical factors here
  - https://www.computer.org/csdl/proceedings-article/qce/2022/911300a238/1IvLLnbwpzi
- Extension to handle heterogenuity here
  - https://ieeexplore.ieee.org/document/9810400
- IEEE JSAC Paper here on log(NM) extension
  - https://ieeexplore.ieee.org/abstract/document/10274098
- Quantum 3D localization here (IPIN presentation)
- Quantum Annealing paper here
  - https://ieeexplore.ieee.org/document/10639748

Thank You ..



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